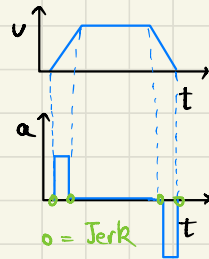



1. DC Motor Model
(exclude phase inductance)
2. Gearbox Model
3. Timing belt drive
4. Power Screw
5. Motion Control
 - A) Trapezoidal Velocity Profile
 - B) S Curve velocity Profile
6. PID Controller
 - A) Linear
 - Mobile Robot
 - H-Plane 3D Printer
 - B) Non Linear
 - Inverted Pendulum
 - Polar Robot

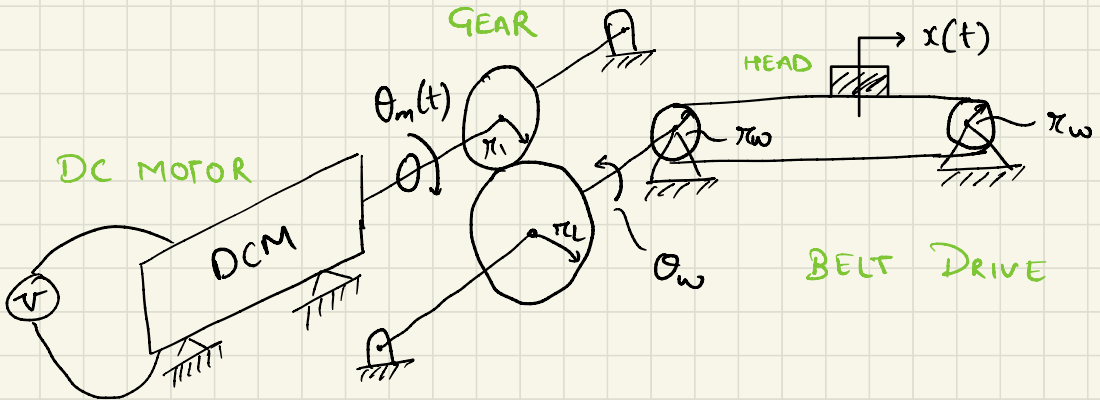


After smoothening corners
⇓



Extra : AC Servo
Stepper
BLDC

CONVEYER SYSTEM



OUTPUT = motion of box

INPUT = voltage to DCM

$$\theta_m r_1 = \theta_w r_2 \quad (\text{no slip})$$

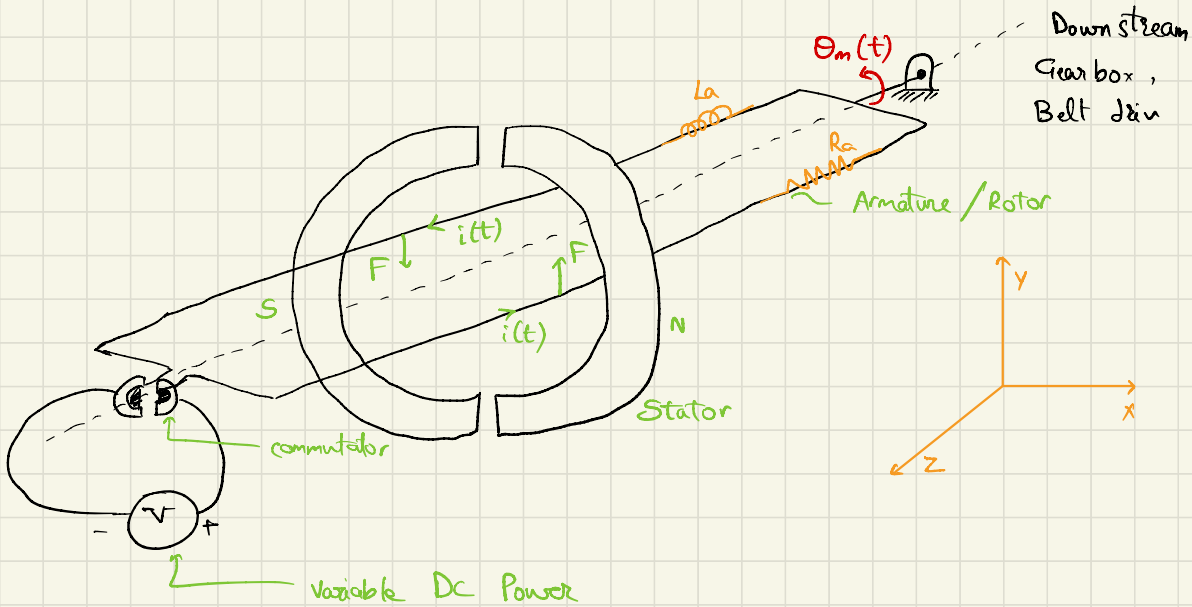
$$x(t) = \theta_w r_1$$

└ radians

$$x(t) = \theta_m \frac{r_1}{r_2} r_1$$

Objective :- Relation b/w voltage and displacement

DC MOTOR



Force given by Fleming's Left Hand Rule (FBI)

EOM :-

(electrical eqⁿ)

$$v(t) = L_a \frac{di_a}{dt} + R_a i_a + e_b(t)$$

(back emf)

$$e_b = k_b \dot{\theta}_m$$

↓
back emf constant

J_m = Rotor Mass MOI about the axis of rotation

D_m = Torsional Damping Nm s/rad
(Loss of energy in bearings)

(mechanical eqⁿ)

$$J_m \ddot{\theta}_m + D_m \dot{\theta}_m = t_m(t) \quad (\text{torque created by motor})$$

$L_a \ll 1$, neglect L_a

$$t_m(t) = K_T i_a$$

↓
Torque constant

from electrical eqⁿ,

$$i_a(t) = \frac{t_m(t)}{K_T}$$

$$R_a i_a + K_b \dot{\theta}_m = v(t)$$

$$\frac{R_a t_m}{K_T} + K_b \dot{\theta}_m = v(t)$$

$$\frac{R_a}{K_T} [J_m \ddot{\theta}_m + D_m \dot{\theta}_m] + K_b \dot{\theta}_m = v(t)$$

$$J_m \ddot{\theta}_m + D_m \dot{\theta}_m + \frac{K_b K_T}{R_a} \dot{\theta}_m = \frac{K_b K_T}{R_a} v(t)$$

→ Take Laplace Transform,

$$s^2 \frac{R_a}{K_T} J_m \theta_m(s) + s D_m \frac{R_a}{K_T} \theta_m(s) + s K_b \theta_m(s) = v(s)$$

↑
uppercase θ_m

$$\theta_m(s) = \frac{1}{\frac{R_a}{K_T} J_m s^2 + \left(D_m \frac{R_a}{K_T} + K_b \right) s} v(s)$$

$$v(s) = \mathcal{L}[v(t)]$$

$$\mathcal{L}[v_0] = \frac{v_0}{s}$$

$$\Theta_m(s) = \frac{1}{s^2 + \alpha s} \frac{v_i}{s}$$

$$\text{where } \alpha = \frac{D_m \frac{R_a}{K_T} + k_b}{\frac{J_m R_a}{K_T}}$$

$$v_i = \frac{K_T}{J_m R_a} v_0$$

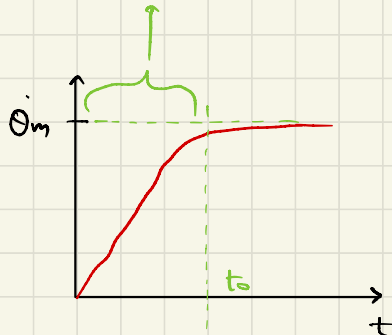
$$\Theta_m(s) = \frac{v_i}{s^3 + \alpha s^2} = \frac{A_1}{s + \alpha} + \frac{B_1}{s^2} + \frac{C_1}{s}$$

$$\Theta_m(t) = A_1 e^{-\alpha t} + B_1 \pi(t) + C_1 u(t)$$

$$\dot{\Theta}_m = -\alpha A_1 e^{-\alpha t} + B_1 u(t)$$

so when $t \rightarrow \infty$,

$$-\alpha A_1 e^{-\alpha t} = 0$$



Characteristic Curve of DC Motor

$$J_m \ddot{\theta}_m + D_m \dot{\theta}_m + \frac{K_b K_T}{R_a} \dot{\theta}_m = \frac{K_b K_T}{R_a} V(t)$$

$$\frac{R_a}{K_T} t_m(t) + K_b \dot{\theta}_m = V(t)$$

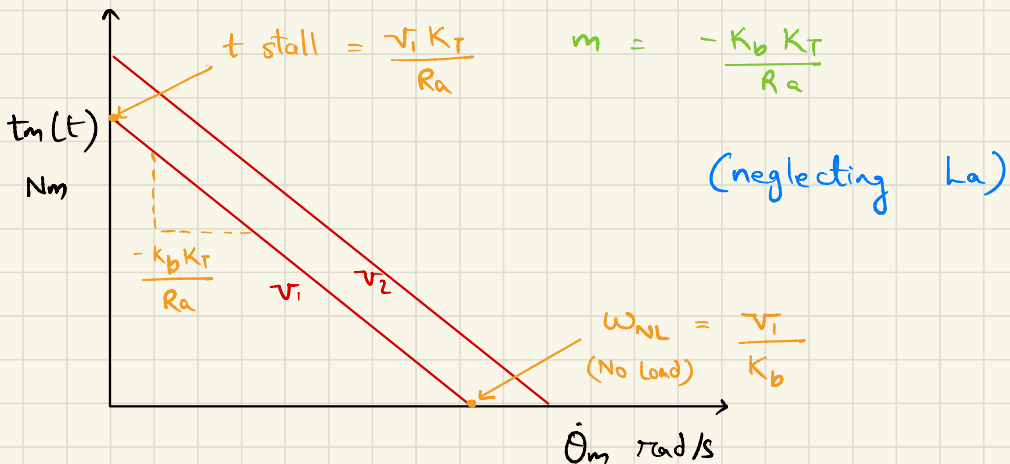
$$t_m(t) \dot{\theta}_m(t) = \text{Power} = \text{Constant}$$

$$y = mx + c$$

$\downarrow$ $\downarrow$
 t_m $\dot{\theta}_m$

$$t_m = \left(\overset{\text{slope}}{-\frac{K_b K_T}{R_a}} \right) \dot{\theta}_m + \left(\overset{\text{y-intercept}}{V(t) \frac{K_T}{R_a}} \right)$$

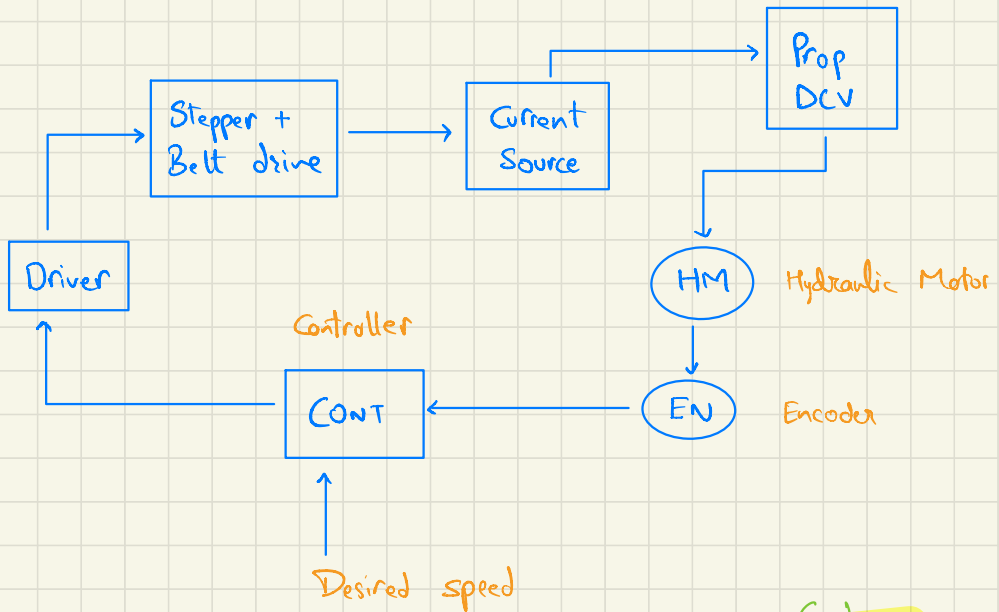
$\downarrow$



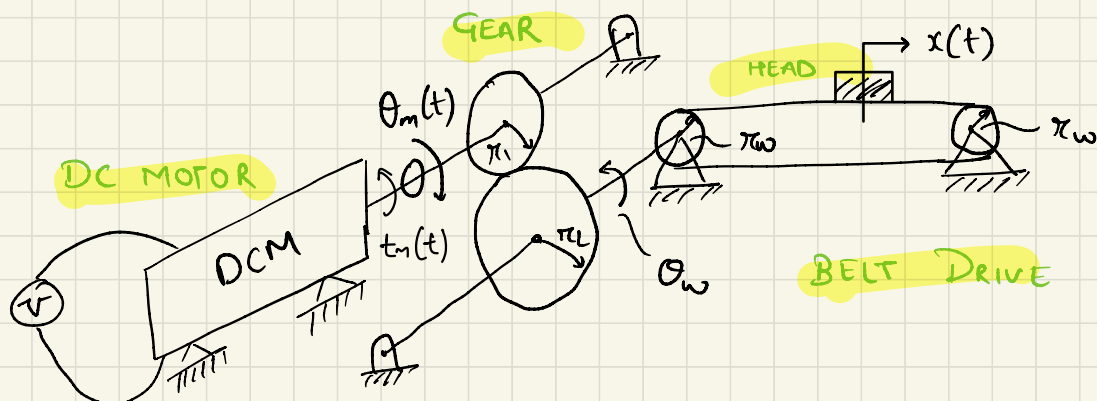
LAB - II

Speed control of Hydraulic motor via positional control of stepper motor timing belt drive assembly.

Schematic :-



CodeSys



From DC Motor :-

$$\frac{R_a \tau_m(t)}{K_T} + K_b \dot{\theta}_m = v(t)$$

$$\cancel{I_a} \frac{d i_a}{d t}$$

$$\tau_m(t) = K_T i_a$$

$$\theta_2 r_2 = \theta_1 r_1$$

($\therefore$ Kinematic)

$$\theta_1 = \theta_m$$

$$x(t) = r_w \theta_2(t)$$

Forward Kinematic :-
relation b/w workspace and joint variable

$$x(t) = r_w \frac{r_1}{r_2} \theta_1$$

FK

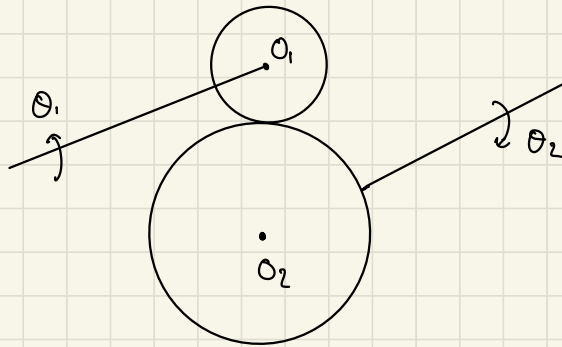
(Forward Kinematic)

$$\theta_1(t) = \frac{\pi_2}{\pi_1 \pi_\omega} x(t)$$

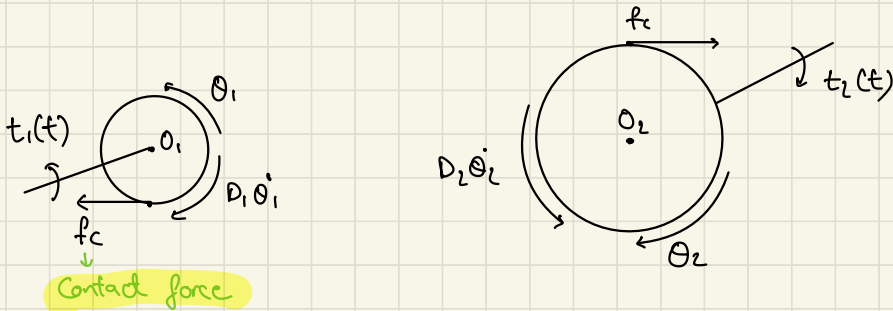
IK

(Inverse Kinematic)

Gear box



FBD



$$\Sigma M_{O_1} = I_{O_1} \ddot{\theta}_1$$

$$\Sigma M_{O_2} = I_{O_2} \ddot{\theta}_2$$

$$I_{O_1} \ddot{\theta}_1 = t_1(t) - D_1 \dot{\theta}_1 - f_c r_1$$

$$-I_{O_2} \ddot{\theta}_2 = -t_2(t) + D_2 \dot{\theta}_2 - f_c r_2$$

$$r_1 \theta_1 = r_2 \theta_2 \quad (\text{No Slip})$$

Write in terms of f_c

$$f_c = \frac{1}{r_2} [I_{O_2} \ddot{\theta}_2 + D_2 \dot{\theta}_2 - t_2(t)]$$

Substituting in θ_1

$$I_{O_1} \ddot{\theta}_1 + D_1 \dot{\theta}_1 + \frac{r_1}{r_2} [I_{O_2} \ddot{\theta}_2 + D_2 \dot{\theta}_2 - t_2(t)] = t_1(t)$$

$$\left[I_{O_1} + \left(\frac{r_1}{r_2} \right)^2 I_{O_2} \right] \ddot{\theta}_1 + \left[D_1 + \left(\frac{r_1}{r_2} \right)^2 D_2 \right] \dot{\theta}_1 = t_1(t) +$$

↑
EOM

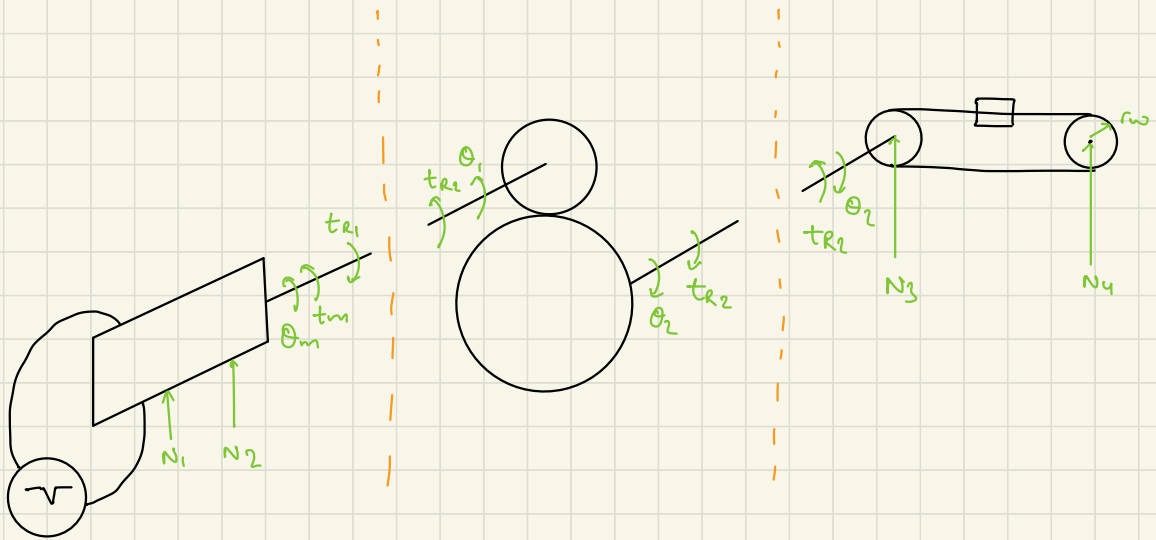
$$+ \frac{r_1}{r_2} t_2(t)$$

Replace I_{01} as J_{01} &

I_{02} as J_{02}

$$\left[J_{01} + \left(\frac{r_1}{r_2} \right)^2 J_{02} \right] \ddot{\theta}_1 + \left[D_1 + \left(\frac{r_1}{r_2} \right)^2 D_2 \right] \dot{\theta}_1 = \tau_1(t) + \frac{r_1}{r_2} \tau_2(t)$$

Separating the system into III components



Motor :-

$$R_m \frac{t_m}{K_T} + K_b \dot{\theta}_m = V(t)$$

$$J_m \ddot{\theta}_m + D_m \dot{\theta}_m = t_m(t) - t_{r1}(t)$$

Gear box :-

$$\begin{aligned} & \left[J_{o1} + \left(\frac{r_1}{r_2} \right)^2 J_{o2} \right] \ddot{\theta}_1 \quad \xrightarrow{\overline{J_{o2}}} \\ & + \left[D_1 + \left(\frac{r_1}{r_2} \right)^2 D_2 \right] \dot{\theta}_1 \quad \xrightarrow{\overline{D_2}} \end{aligned} = \begin{aligned} & t_{r1}(t) \\ & + \\ & \frac{r_1}{r_2} t_{r2}(t) \end{aligned}$$

Belt drive :-

$$- \left[J_{O_3} + J_{O_4} + m r \omega^2 \right] \ddot{\theta}_2 - \left[D_3 + D_4 \right] \dot{\theta}_2 = t_{R_2}(t)$$

$\hookrightarrow \bar{J}_{34}$ $\hookrightarrow \bar{D}_{34}$

$$- \bar{J}_{34} \ddot{\theta}_2 - \bar{D}_{34} \dot{\theta}_2 = t_{R_2}(t)$$

Putting in Gearbox,

$$\left[\bar{J}_2 + \bar{J}_{34} \left(\frac{r_1}{r_2} \right)^2 \right] \ddot{\theta}_1 + \left[\bar{D}_2 + \bar{D}_{34} \left(\frac{r_1}{r_2} \right)^2 \right] \dot{\theta}_1 = t_{R_1}$$

Combine everything

$$\frac{R_a t_m(t)}{K_T} + K_b \dot{\theta}_m = v(t)$$

Adding gearbox + motor,
($\theta_m = \theta_1$)

$$\left[J_m + \bar{J}_2 + \left(\frac{r_1}{r_2} \right)^2 \bar{J}_{34} \right] \ddot{\theta}_m +$$

$= t_m(t)$

$$\left[D_m + \bar{D}_2 + \bar{D}_{34} \left(\frac{r_1}{r_2} \right)^2 \right] \dot{\theta}_m$$

Substitute this $t_m(t)$
to motor.

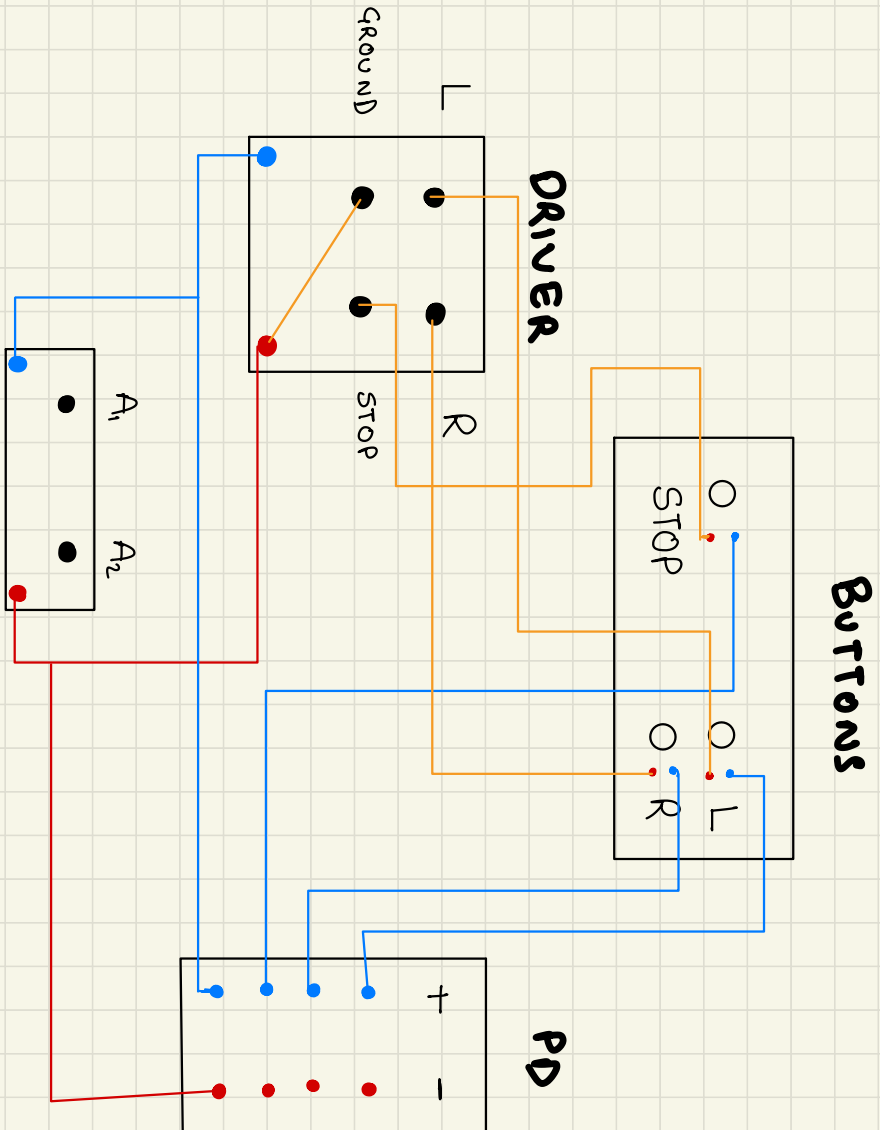
PID controller,

$$\tau(t) = -K_p (\theta_m - \theta_{md}) - K_D (\dot{\theta}_m - \dot{\theta}_{md}) \\ - K_I \int_0^t (\theta_m(\tau) - \theta_{md}(\tau)) d\tau$$

$\theta_{md} \rightarrow$ TVP
(Trapezoidal Velocity profile)

LAB - III

MANUAL



PLC

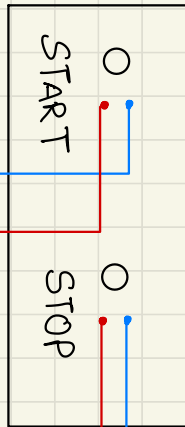
Inputs

$\%I \times 0.0$ Start
 $\%I \times 0.1$ Stop
 $\%I \times 0.2$ A1
 $\%I \times 0.3$ A2

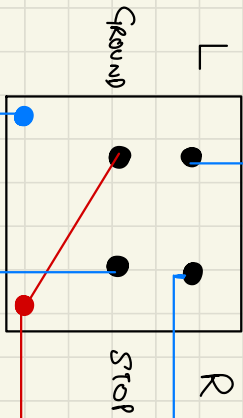
Outputs

Stop $\%Q \times 0.0$
 Left $\%Q \times 0.1$
 Right $\%Q \times 0.2$
 Stop-OUT_R $\%Q \times 0.3$
Main Stop $\%Q \times 1.0$

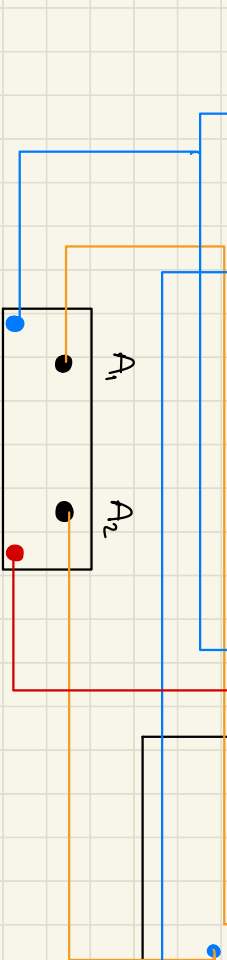
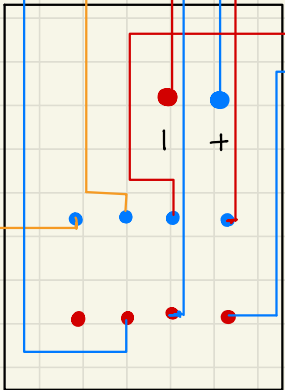
Buttons



DRIVER

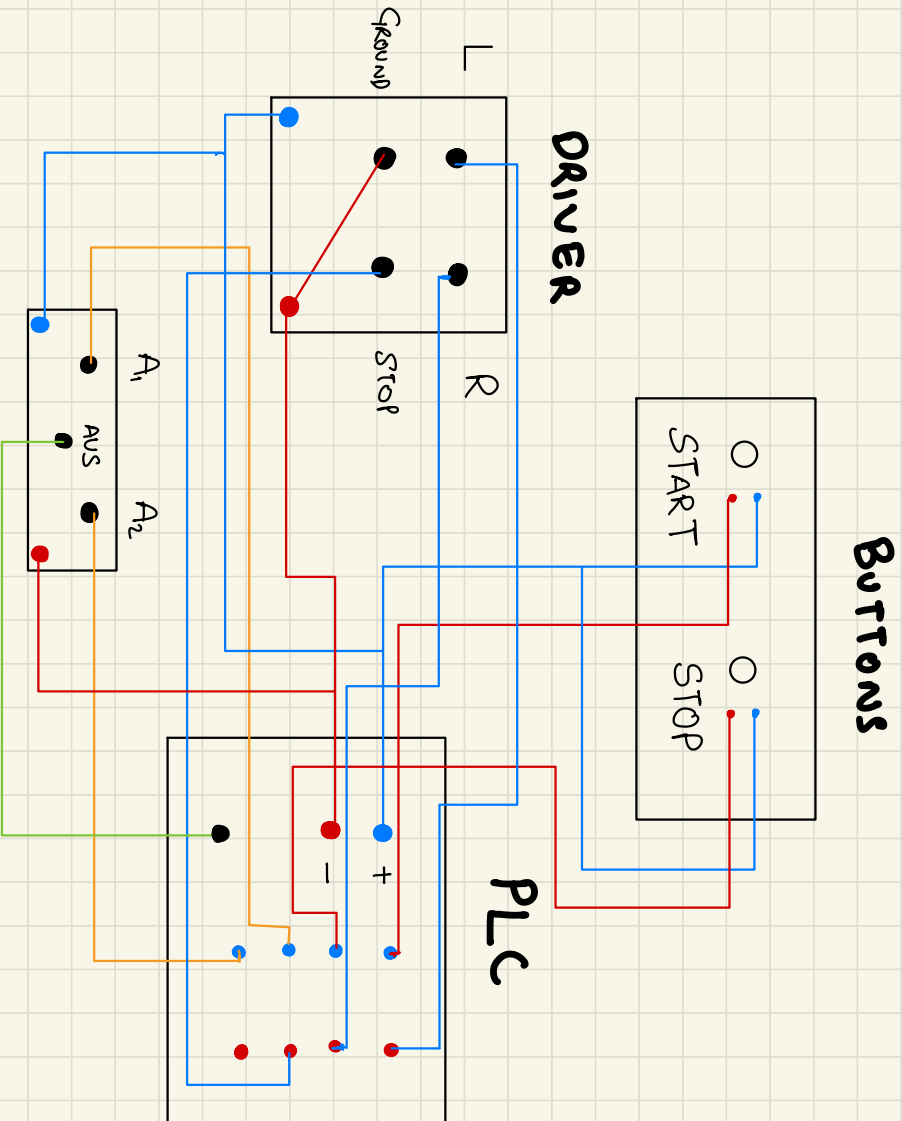


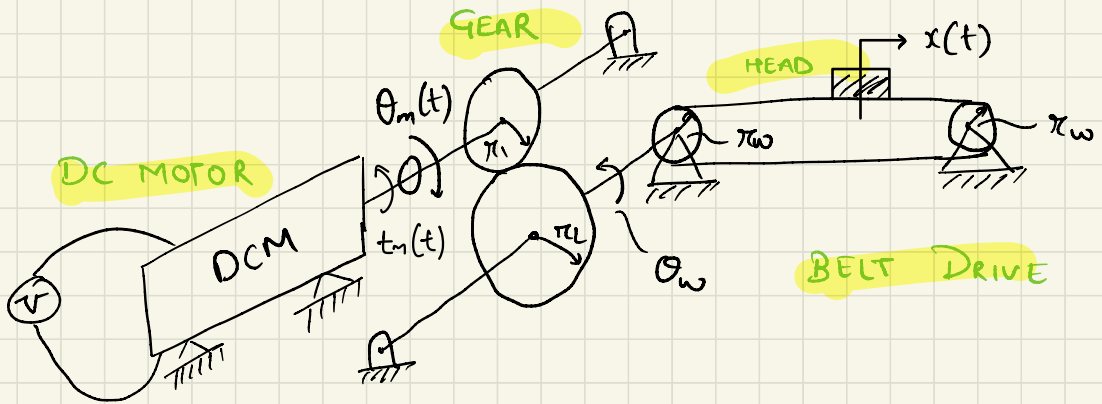
PLC



PLC

With Analog Input





General equation :-

$$\bar{J}\ddot{\theta} + \bar{D}\dot{\theta} = \tau(t)$$

Open loop
(no encoder)

closed loop

$\theta_d(t)$ = Desired Trajectory

Open Loop :-

$$\bar{J}\ddot{\theta} + \bar{D}\dot{\theta} = \tau(t) \quad \rightarrow (1)$$

$$\tau(t) = \bar{J}\ddot{\theta}_d + \bar{D}\dot{\theta}_d \quad \rightarrow (2)$$

computed torque controller

$$\bar{J}\ddot{\theta} + \bar{D}\dot{\theta} = \bar{J}\ddot{\theta}_d + \bar{D}\dot{\theta}_d$$

$$\bar{D}(\ddot{\theta} - \ddot{\theta}_d) + \bar{D}(\dot{\theta} - \dot{\theta}_d) = 0$$

$$e(t) = \theta(t) - \theta_d(t)$$

$$\bar{D} \ddot{e} + \bar{D} \dot{e} = 0$$

$$\begin{array}{l} e(t) \rightarrow 0 \\ e(0) = 0 \end{array} \quad , \quad \begin{array}{l} t \rightarrow \infty \\ \dot{e}(0) = 0 \end{array}$$

Problem of open loop ,

- unable to change the eigenstructure.
- no poles in the LHP which leads to instability.

Closed Loop :-

PID Controller

$$\bar{J}\ddot{\theta} + \bar{D}\dot{\theta} = \tau(t)$$

$$\tau(t) = \bar{J} \left[-k_p (\theta - \theta_d) - k_d (\dot{\theta} - \dot{\theta}_d) - k_I \int_0^t [\theta(\tau) - \theta_d(\tau)] d\tau \right] + \bar{D}\dot{\theta} + \bar{J}\ddot{\theta}_d$$

→ Computed torque control

$$\bar{J} \left[(\ddot{\theta} - \ddot{\theta}_d) + k_d (\dot{\theta} - \dot{\theta}_d) + k_p (\theta - \theta_d) + k_I \int_0^t [\theta(\tau) - \theta_d(\tau)] d\tau \right] = 0$$

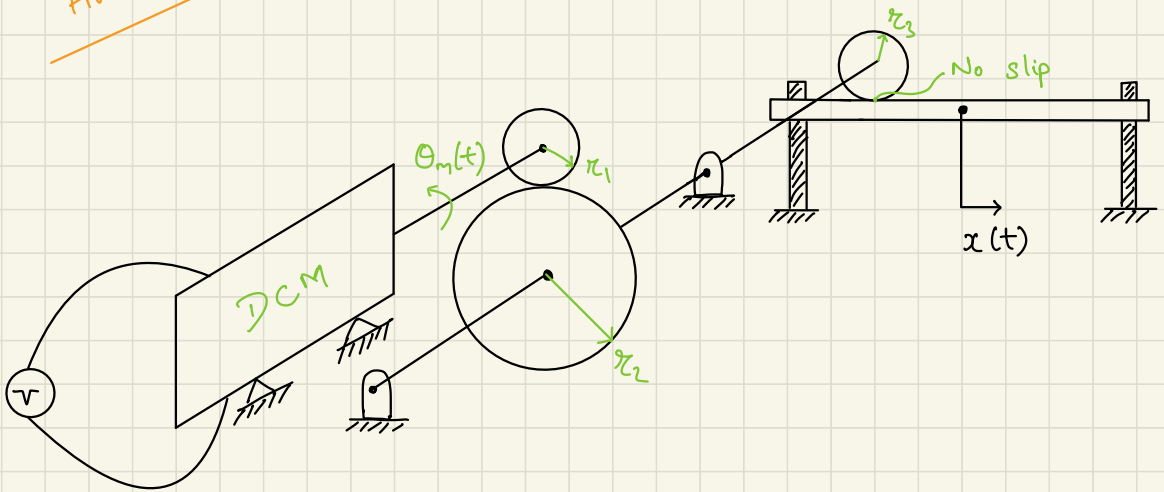
(Differentiating one more time)

$$\ddot{e} + k_d \dot{e} + k_p e + k_I e = 0$$

$$\lambda^3 + k_d \lambda^2 + k_p \lambda + k_I = 0$$

$$\lambda^3 + K_D \lambda^2 + K_P \lambda + K_I = (\lambda - \lambda_1) (\lambda - \lambda_2) (\lambda - \lambda_3)$$

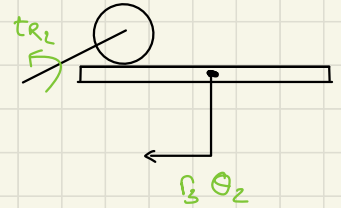
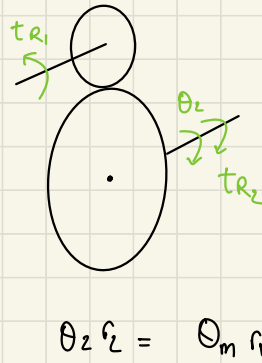
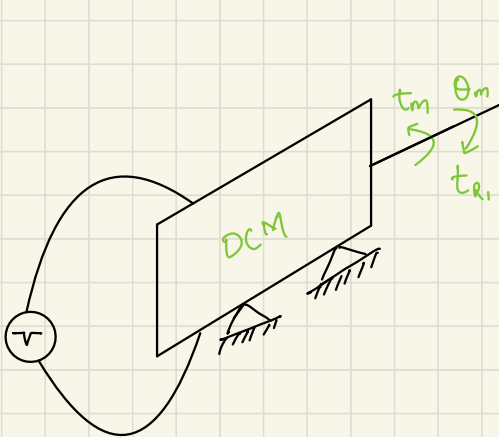
HW-1



Advantages :-

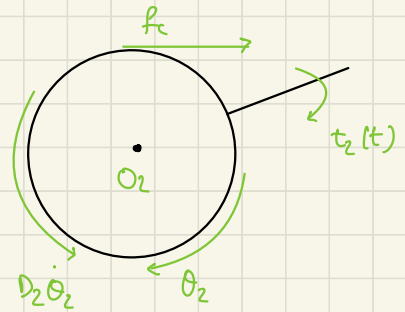
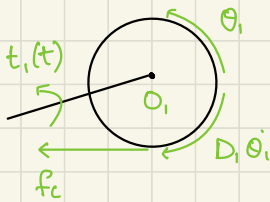
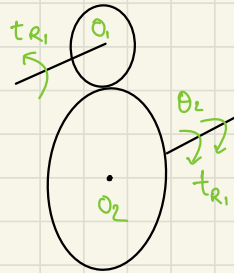
- i) Mitigation of downstream inertia and damping constants
- ii) Increase in resolution (No slip, $\theta_2 = \frac{r_1}{r_2} \theta_1$)

FBD

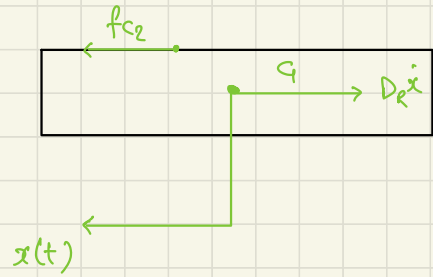
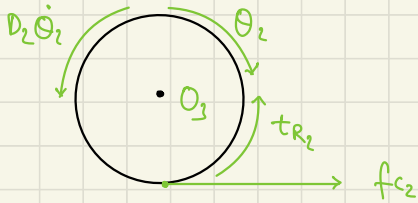


$$x(t) = \frac{r_1}{r_2} \theta_2(t)$$

Gear box



Rack & Pinion



EOM

Motor :-

(electrical eqⁿ)

$$v(t) = L_a \frac{di_a}{dt} + R_a i_a + K_b \dot{\theta}_m$$

(mechanical eqⁿ)

$$J_m \ddot{\theta}_m + D_m \dot{\theta}_m = t_m(t)$$

$$R_a i_a + K_b \dot{\theta}_m = v(t)$$

$$\frac{R_a t_m}{K_T} + K_b \dot{\theta}_m = v(t)$$

$$\frac{R_a}{K_T} (J_m \ddot{\theta}_m + D_m \dot{\theta}_m) + K_b \dot{\theta}_m = v(t)$$

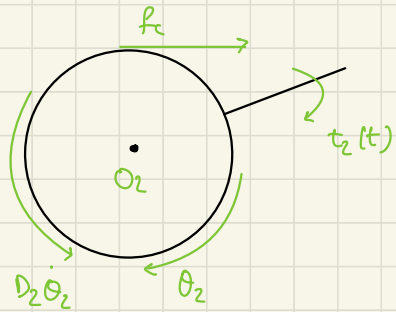
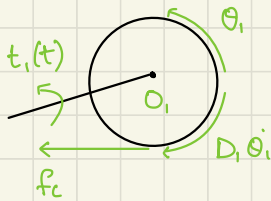
$$J_m \ddot{\theta}_m + D_m \dot{\theta}_m + \frac{K_b K_T}{R_a} \dot{\theta}_m = \frac{K_b K_T}{R_a} v(t)$$

$$\therefore J_m = J_o, \quad D_m = D_o, \quad \theta_m = \theta_1$$

$$J_o \ddot{\theta}_1 + D_o \dot{\theta}_1 + \frac{K_b K_T}{R_a} \dot{\theta}_1 = \frac{K_b K_T}{R_a} v(t)$$

$$J_o \ddot{\theta}_1 + D_o \dot{\theta}_1 = t_m(t) - t_{e1}(t)$$

Gearbox



$$\sum M_{O_1} = I_{O_1} \ddot{\theta}_1$$

$$\sum M_{O_2} = I_{O_2} \ddot{\theta}_2$$

$$I_{O_1} \ddot{\theta}_1 = t_1(t) - D_1 \dot{\theta}_1 - f_c r_1$$

$$-I_{O_2} \ddot{\theta}_2 = -t_2(t) + D_2 \dot{\theta}_2 - f_c r_2$$

According no slip,

$$r_1 \theta_1 = r_2 \theta_2$$

$$f_2 = \frac{1}{r_2} [I_{O_1} \ddot{\theta}_1 + D_2 \dot{\theta}_1 - t_1(t)]$$

Substituting in θ_1

$$I_{o1} \ddot{\theta}_1 + D_1 \dot{\theta}_1 + \frac{r_1}{r_2} \left[I_{o2} \ddot{\theta}_2 + D_2 \dot{\theta}_2 - t_2(t) \right] = t_1(t)$$

$$\left[I_{o1} + \left(\frac{r_1}{r_2} \right)^2 I_{o2} \right] \ddot{\theta}_1 + \left[D_1 + \left(\frac{r_1}{r_2} \right)^2 D_2 \right] \dot{\theta}_1 = t_1(t) + \frac{r_1}{r_2} t_2(t)$$

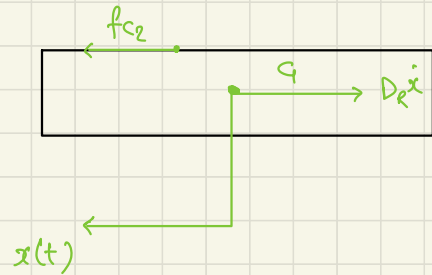
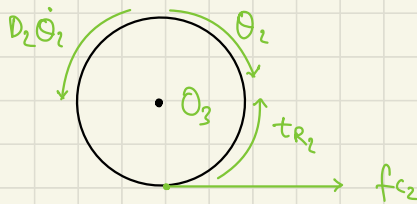


EOM

Replace I_{o1} as J_{o1} &
 I_{o2} as J_{o2}

$$\left[J_{o1} + \left(\frac{r_1}{r_2} \right)^2 J_{o2} \right] \ddot{\theta}_1 + \left[D_1 + \left(\frac{r_1}{r_2} \right)^2 D_2 \right] \dot{\theta}_1 = t_1(t) + \frac{r_1}{r_2} t_2(t)$$

Rack & Pinion



$$\sum M_{O_3} = -J_{O_3} \ddot{\theta}_2 = r_3 f_{c2} + D_3 \dot{\theta}_2 + t_{R2}(t)$$

$$\sum F_x = -m_R \ddot{x} = -f_{c2} + D_R \dot{x}$$

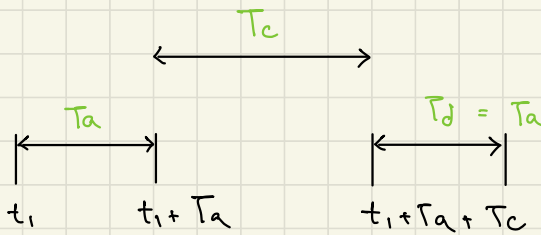
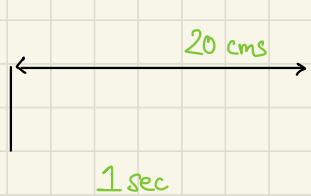
$$f_{c2} = m_R \ddot{x} + D_R \dot{x}$$

$$t_{R2}(t) = - \left[J_{O_3} \ddot{\theta}_2 + D_3 \dot{\theta}_2 + m_R r_3^2 \ddot{\theta}_2 + D_R r_3^2 \dot{\theta}_2 \right]$$

Combining all eq's,

$$\bar{J}\ddot{\theta}_m + \bar{D}\dot{\theta}_m = v(t)$$

$$v(t) = \bar{J}\ddot{\theta}_m$$



$$t_1 < t \leq t_1 + T_a$$

$$t_1 + T_a < t \leq t_1 + T_a + T_c$$

$$t_1 + T_a + T_c \leq t \leq t_2$$

$$x_d(t) = MCO1$$

$$\theta_d(t) = \frac{\pi_2}{\pi_1 \pi_3} x_d(t)$$

$$T = 2T_a + T_c$$

$$S = 2S_a + S_c$$

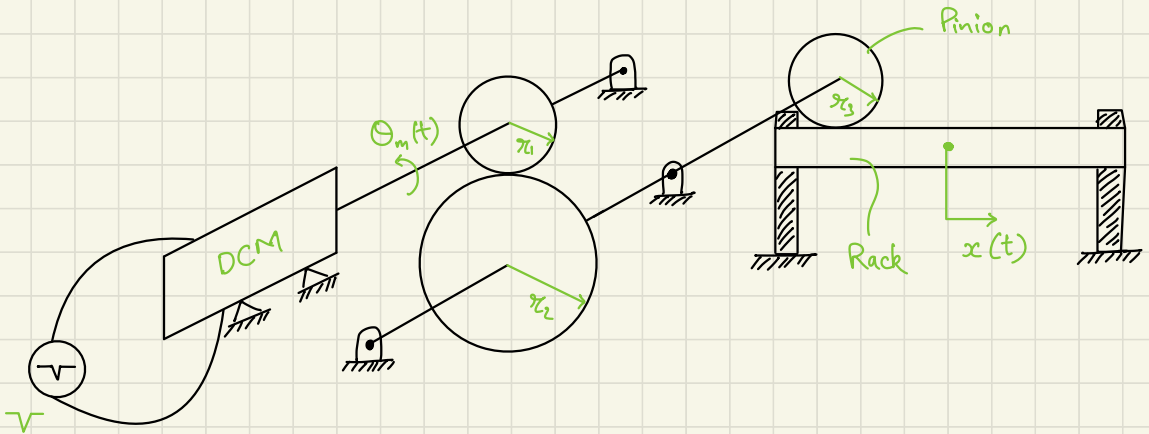
$$\tau(t) = \bar{J} \ddot{\theta}_d + \bar{D} \dot{\theta}_m - \bar{J} \left[k_p (\theta_m - \theta_d) + k_d (\dot{\theta}_m - \dot{\theta}_d) + k_I \int_0^t (\theta_m(\tau) - \theta_d(\tau)) d\tau \right]$$

$$e(t) = \theta_m - \theta_d$$

$$\bar{J} \left(\ddot{e} + k_d \dot{e} + k_p e + k_I \int_0^t e(\tau) d\tau \right) = 0$$

$$x(t) = \pi_3 \frac{\pi_1}{\pi_2} \theta_m(t)$$

Q1.



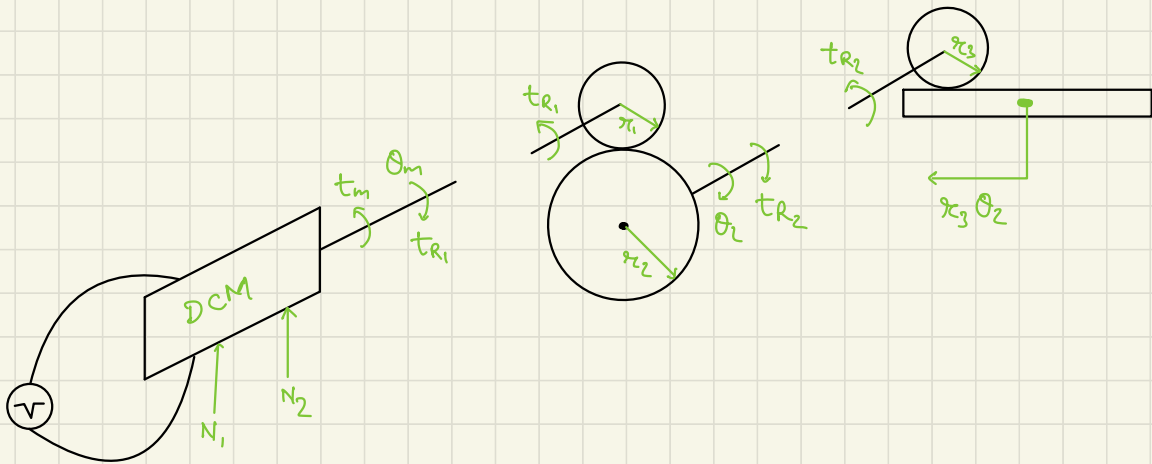
- Derive the kinematic relationship b/w $x(t)$ and $\theta_m(t)$
- Draw an FBD of the assembly showing the reactionary forces and torques
- Derive the EOM relating the voltage $v(t)$ to the rack displacement $x(t)$
- Assume the EOM is of the form
$$\ddot{\theta}_m + \bar{D}\dot{\theta}_m = v(t)$$

Design a computed torque PD controller to place the poles @ -40 , -50 .

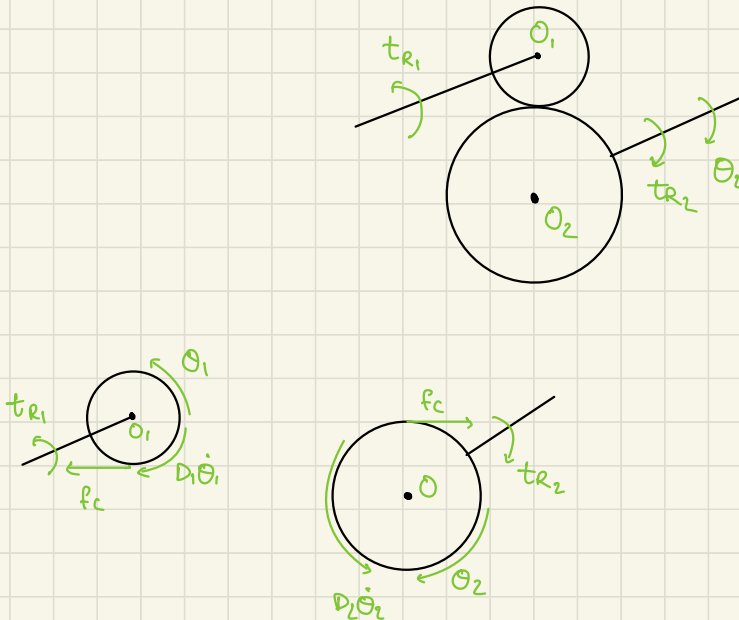
- The rack has to be moved 50 cms in 3 seconds. Design the TVP for the motion

B.

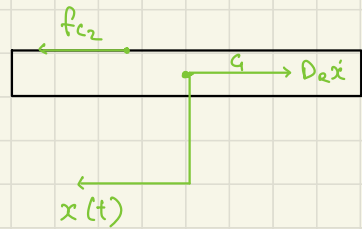
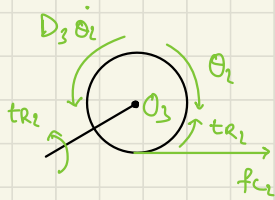
Free Body Diagrams :-



Gearbox :-



Rack and Pinion



C.

Motor :-

(electrical eqⁿ)
$$v(t) = L_a \frac{di_a}{dt} + R_a i_a + k_b \dot{\theta}_m$$

$$R_a i_a + k_b \dot{\theta}_m = v(t)$$

$$\frac{R_a t_m}{K_r} + k_b \dot{\theta}_m = v(t) \quad (\because t_m = K_r i_a)$$

①

(mechanical eqⁿ)
$$J_{O_1} \ddot{\theta}_1 + D_{O_1} \dot{\theta}_1 = t_m$$

Substitute in ①

$$\frac{R_a}{K_r} (J_{O_1} \ddot{\theta}_1 + D_{O_1} \dot{\theta}_1) + k_b \dot{\theta}_m = v(t)$$

$$J_{O_1} \ddot{\theta}_1 + D_{O_1} \dot{\theta}_1 + \frac{k_b K_r}{R_a} \dot{\theta}_1 = K_b K_r v(t)$$

$$J_{O_1} \ddot{\theta}_1 + D_{O_1} \dot{\theta}_1 = K_b K_r v(t) - \frac{K_b K_r}{R_a} \theta_1$$

$$J_{O_1} \ddot{\theta}_1 + D_{O_1} \dot{\theta}_1 = t_m(t) - t_{R_1}(t) \quad \text{---} \quad (2)$$

Gearbox :-

$$\Sigma M_{O_1} = J_{O_1} \ddot{\theta}_1$$

$$J_{O_1} \ddot{\theta}_1 = t_{R_1} - f_{c_1} \pi_1 - D_1 \dot{\theta}_1$$

$$t_{R_1} = J_{O_1} \ddot{\theta}_1 + f_{c_1} \pi_1 + D_1 \dot{\theta}_1 \quad \text{---} \quad (3)$$

$$\Sigma M_{O_2} = -J_{O_2} \ddot{\theta}_2$$

$$-J_{O_2} \ddot{\theta}_2 = D_2 \dot{\theta}_2 - f_{c_1} \pi_2 - t_{R_2}$$

$$f_{c_1} \pi_2 = J_{O_2} \ddot{\theta}_2 + D_2 \dot{\theta}_2 - t_{R_2}$$

$$f_{c_1} = (J_{O_2} \ddot{\theta}_2 + D_2 \dot{\theta}_2 - t_{R_2}) \frac{1}{\pi_2} \quad \text{---} \quad (4)$$

Rack and Pinion :-

$$\Sigma M_{O_3} = -J_{O_3} \ddot{\theta}_2$$

$$-J_{O_3} \ddot{\theta}_2 = \pi_3 f_{c_2} + D_3 \dot{\theta}_2 + t_{R_2}$$

$$t_{R_2} = -J_{O_3} \ddot{\theta}_2 - \pi_3 f_{c_2} - D_3 \dot{\theta}_2 \quad \text{---} \quad (5)$$

$$\sum F_x = -m_R \ddot{x}$$

$$-m_R \ddot{x} = -f_{c2} + D_R \dot{x}$$

$$f_{c2} = m_R \ddot{x} + D_R \dot{x}$$

Substituting in (5)

$$t_{R2} = -J_{O3} \ddot{\theta}_1 - \pi_3 m_R \ddot{x} - \pi_3 D_R \dot{x} - D_3 \dot{\theta}_2 \quad \longrightarrow \quad (6)$$

Combine everything,

Put (6) $\rightarrow$ (4),

$$f_{c1} = \left(J_{O2} \ddot{\theta}_1 + D_2 \dot{\theta}_1 + J_{O3} \ddot{\theta}_1 + \pi_3 m_R \ddot{x} + \pi_3 D_R \dot{x} + D_3 \dot{\theta}_2 \right) \frac{1}{\pi_2} \quad \longrightarrow \quad (7)$$

Put (7) $\rightarrow$ (3),

$$t_{R1} = J_{O1} \ddot{\theta}_1 + D_1 \dot{\theta}_1 + \frac{\pi_1}{\pi_2} \left(J_{O2} \ddot{\theta}_1 + D_2 \dot{\theta}_1 + J_{O3} \ddot{\theta}_1 + \pi_3 m_R \ddot{x} + \pi_3 D_R \dot{x} + D_3 \dot{\theta}_2 \right) \quad \longrightarrow \quad (8)$$

Put ⑧ $\rightarrow$ ②,

$$J_0 \ddot{\theta}_1 + D_1 \dot{\theta}_1 = t_m(t) - J_0 \ddot{\theta}_1 - D_1 \dot{\theta}_1$$

$$- \frac{\pi_1}{\pi_2} \left(J_{02} \ddot{\theta}_2 + D_2 \dot{\theta}_2 + J_{03} \ddot{\theta}_2 + \pi_3 m_R \ddot{x} + \pi_3 D_R \dot{x} + D_3 \dot{\theta}_2 \right)$$

$$t_m(t) = 2J_0 \ddot{\theta}_1 + 2D_1 \dot{\theta}_1 + \frac{\pi_1}{\pi_2} \left(J_{02} \ddot{\theta}_2 + D_2 \dot{\theta}_2 + J_{03} \ddot{\theta}_2 + \pi_3 m_R \ddot{x} + \pi_3 D_R \dot{x} + D_3 \dot{\theta}_2 \right)$$

$\longrightarrow$ ⑨

Put ⑨ $\rightarrow$ ①

$$\frac{R_a}{K_T} \left[2J_0 \ddot{\theta}_1 + 2D_1 \dot{\theta}_1 + \frac{\pi_1}{\pi_2} \left(J_{02} \ddot{\theta}_2 + D_2 \dot{\theta}_2 + J_{03} \ddot{\theta}_2 + \pi_3 m_R \ddot{x} + \pi_3 D_R \dot{x} + D_3 \dot{\theta}_2 \right) \right] + K_b \dot{\theta}_m = v(t)$$

A.

$$\theta_2 \pi_2 = \theta_m \pi_1$$

$$x(t) = \pi_3 \theta_2(t)$$

$$x(t) = \frac{\pi_1 \cdot \pi_3}{\pi_2} \cdot \theta_m(t)$$

D.

$$\bar{J}\ddot{\theta} + \bar{D}\dot{\theta} = v(t)$$

PD Controller ,

$$v(t) = \bar{J} \left[-K_p (\theta - \theta_d) - K_d (\dot{\theta} - \dot{\theta}_d) \right] + \bar{D}\dot{\theta} + \bar{J}\ddot{\theta}_d$$

Computed Torque control

$$\bar{J} [(\ddot{\theta} - \ddot{\theta}_d) + K_d (\dot{\theta} - \dot{\theta}_d) + K_p (\theta - \theta_d)] = 0$$

$$\ddot{e} + K_d \dot{e} + K_p e = 0$$

$$\lambda^2 + K_d \lambda + K_p = (\lambda + 40)(\lambda + 50)$$

$$(\lambda + 40)(\lambda + 50) = \lambda^2 + 50\lambda + 40\lambda + 2000$$

$$(\lambda + 40)(\lambda + 50) = \lambda^2 + 90\lambda + 2000$$

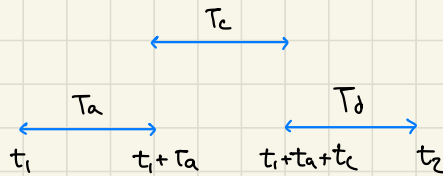
$$\therefore K_D = 90$$

$$K_P = 2000$$

E.

50 cms $\rightarrow$ 3 seconds

For TVP,

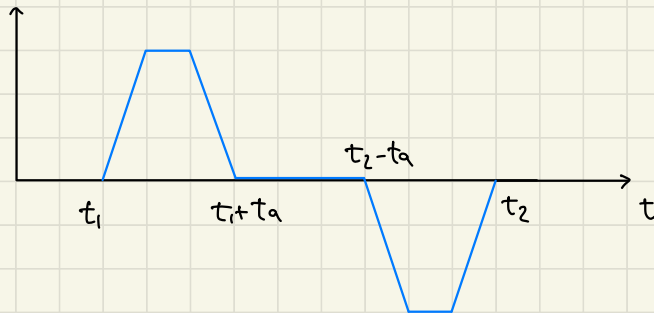


where,

T_a = acceleration

T_c = constant speed

T_d = deceleration



Assume $S_a = 5 \text{ cms}$

$\therefore S_d = S_a = 5 \text{ cms}$

$$S = S_a + S_c + S_d$$

$$S_c = 50 - 10 = 40 \text{ cms}$$

$$T_a = \frac{S_a T}{2 S_a + \frac{S_c}{2}}$$

$$= \frac{5 \times 3}{2(5) + \frac{40}{2}} = \frac{15}{30} = \frac{1}{2}$$

$$T_a = 0.5 \text{ s} = T_d$$

$$T = T_a + T_c + T_d$$

$$T_c = 3 - 1$$

$$T_c = 2 \text{ s}$$

$$\therefore S_a = 5 \text{ cms}$$

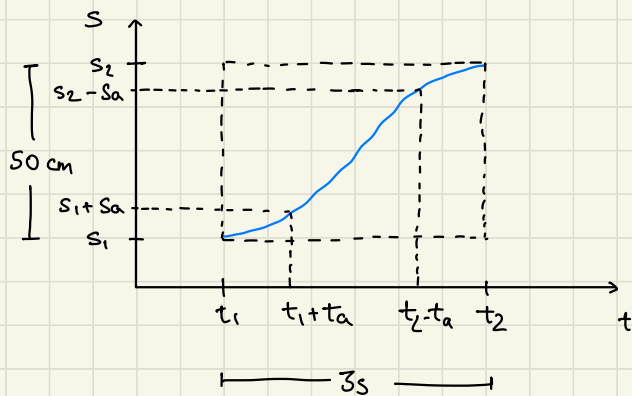
$$T_a = 0.5 \text{ s}$$

$$S_c = 40 \text{ cm}$$

$$T_c = 2 \text{ s}$$

$$S_d = 5 \text{ cms}$$

$$T_d = 0.5 \text{ s}$$

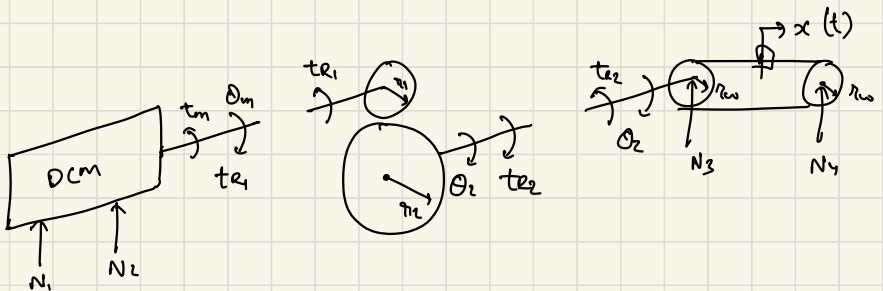
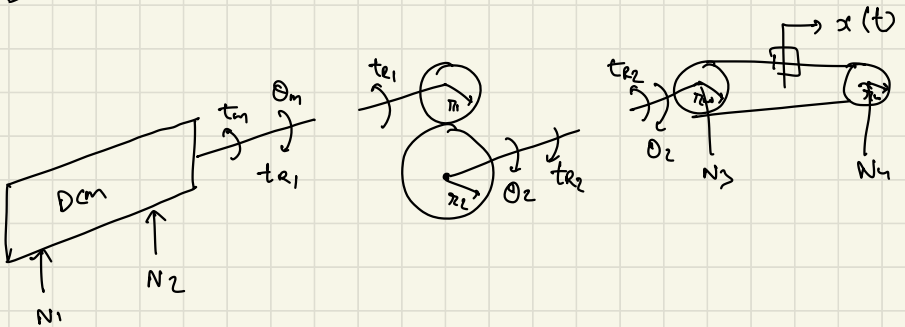
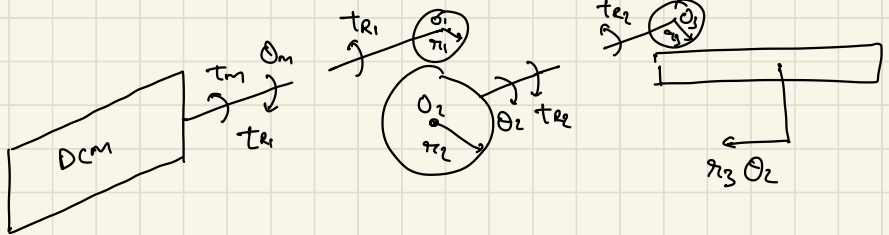
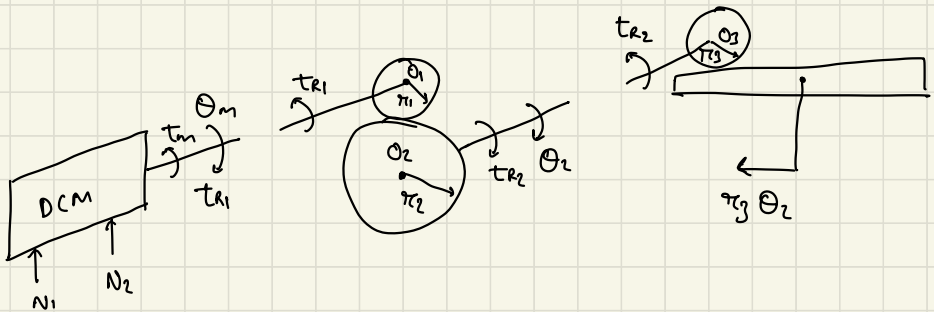


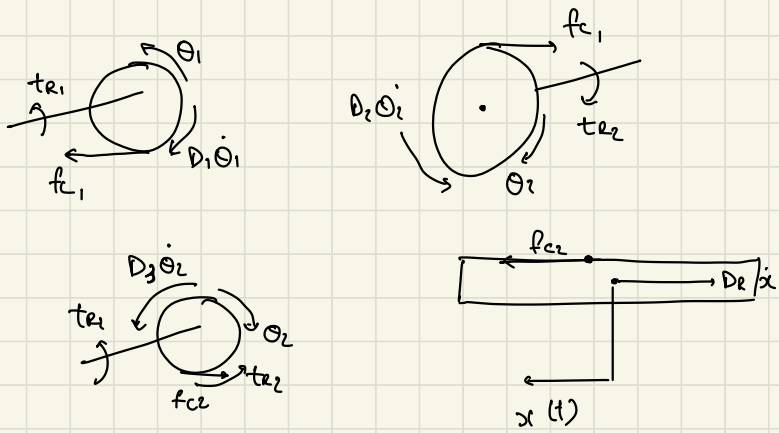
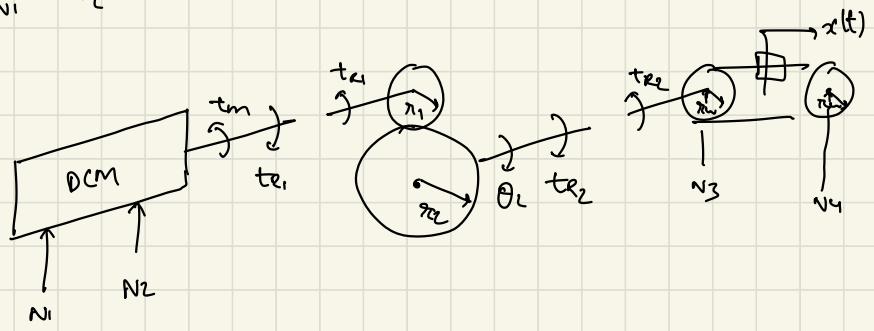
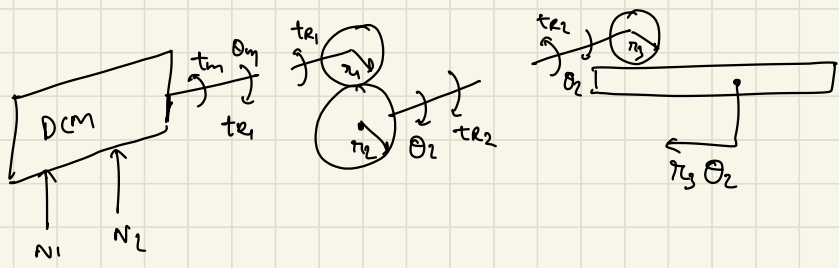
Trapezoidal Velocity Profile

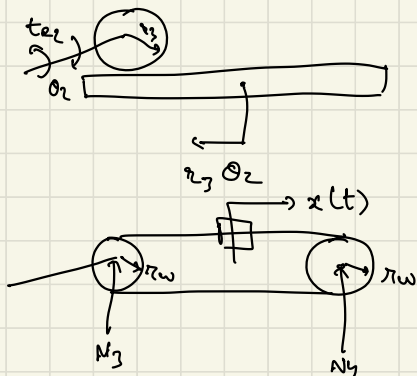
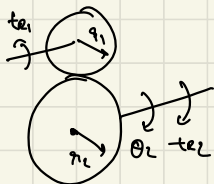
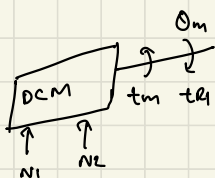
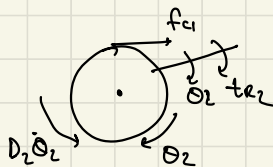
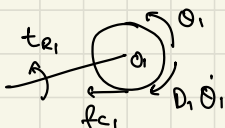
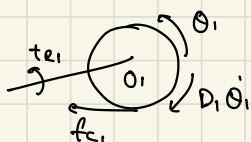
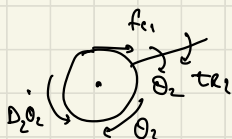
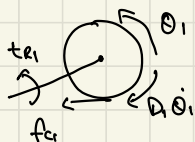
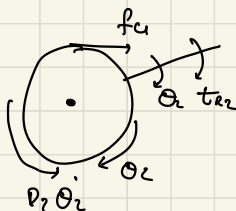
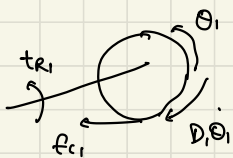
$$\begin{aligned} t_1 &\leq t < t_1 + T_a \\ t_1 + T_a &\leq t < t_1 + T_a + T_c \\ t_2 - T_a &\leq t \leq t_2 \end{aligned}$$

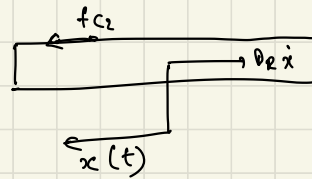
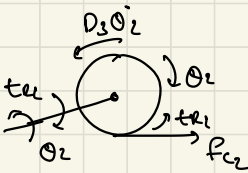
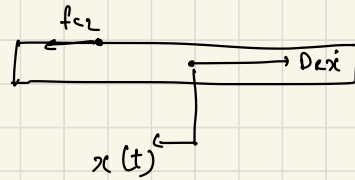
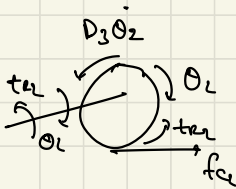
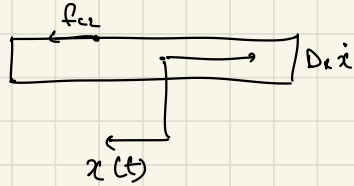
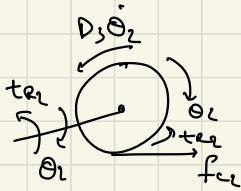
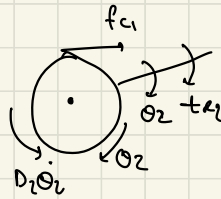
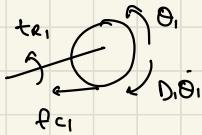
↑ write with values of t_1, t_2, t_3

M-1









Kinematics :-

in back of Pinion

$$\theta_1 r_1 = \theta_2 r_2$$

$$\theta_1 = \theta_m$$

$$\theta_m r_1 = \theta_2 r_2$$

$$x(t) = r_3 \theta_2(t)$$

$$x(t) = \frac{r_1 r_3}{r_2} \theta_m(t)$$

$$\Theta_1 r_1 = \Theta_2 r_2$$

$$\Theta_1 = \Theta_m$$

$$\Theta_m r_1 = \Theta_2 r_2$$

$$x(t) = r_3 \Theta_2(t)$$

$$x(t) = r_3 \frac{r_1}{r_2} \Theta_m(t)$$

$$\Theta_1 r_1 = \Theta_2 r_2$$

$$\Theta_1 = \Theta_m$$

$$\Theta_m r_1 = \Theta_2 r_2$$

$$x(t) = r_3 \Theta_2(t)$$

$$x(t) = r_3 \frac{r_1}{r_2} \Theta_m(t)$$

ii) Belt drive

$$\Theta_1 r_1 = \Theta_2 r_2$$

$$\Theta_1 = \Theta_m$$

$$\Theta_2 = \Theta_w$$

$$\Theta_m r_1 = \Theta_w r_2$$

$$x(t) = \Theta_w r_w$$

$$x(t) = \frac{r_1 - r_w}{r_2} \Theta_m$$

$$\theta_1 r_1 = \theta_2 r_2$$

$$\theta_1 = \theta_m$$

$$\theta_2 = \theta_w$$

$$\theta_m r_1 = \theta_w r_2$$

$$x(t) = r_w \theta_w$$

$$x(t) = r_w \frac{r_1}{r_2} \theta_m$$

$$\theta_1 r_1 = \theta_2 r_2$$

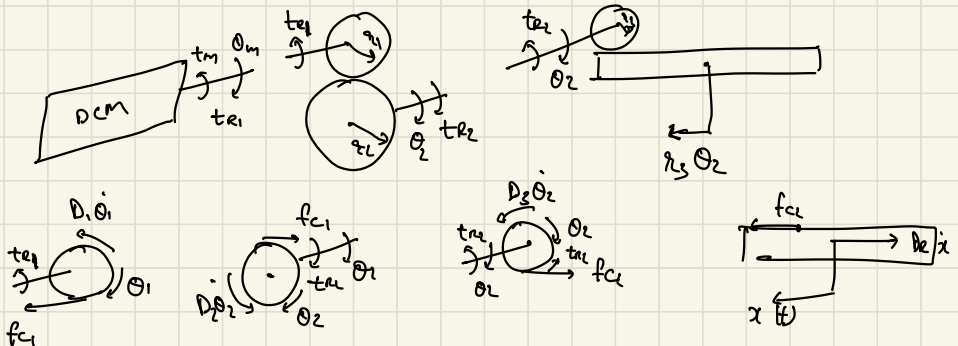
$$\theta_1 = \theta_m$$

$$\theta_2 = \theta_w$$

$$\theta_m r_1 = \theta_w r_2$$

$$x(t) = r_w \theta_w$$

$$x(t) = r_w \frac{r_1}{r_2} \theta_m$$



EOM :-

$$v(t) = \cancel{L_a \frac{di_a}{dt}}^0 + R_a i_a + K_b \dot{\theta}_m$$

$$v(t) = L_a \frac{di_a}{dt} + R_a i_a + K_b \dot{\theta}_m$$

$$v(t) = \cancel{L_a \frac{di_a}{dt}}^0 + R_a i_a + K_b \dot{\theta}_m$$
$$t_m = K_T i_a$$

$$J_{o1} \ddot{\theta}_1 + D_{o1} \dot{\theta}_1 = t_m$$

$$J_{o1} \ddot{\theta}_1 + D_{o1} \dot{\theta}_1 = t_m$$

$$v(t) = L_a \frac{di_a}{dt} + R_a i_a + K_b \dot{\theta}_m$$

$$t_m = K_T i_a$$

$$J_{o1} \ddot{\theta}_1 + D_{o1} \dot{\theta}_1 = t_m$$

$$\frac{R_a}{K_T} (J_{o1} \ddot{\theta}_1 + D_{o1} \dot{\theta}_1) + K_b \dot{\theta}_m = v(t)$$

$$J_{o1} \ddot{\theta}_1 + D_{o1} \dot{\theta}_1 + \frac{K_b K_T}{R_a} \dot{\theta}_m = K_b K_T v(t)$$

$$= t_m(t) - t_{e1}(t)$$

$$t_m(t) - t_{e1}(t)$$

$$t_m(t) - t_{e1}(t)$$

$$t_m(t) - t_{e1}(t)$$

Gearbox :-

$$\Sigma M_{O_1} = J_{O_1} \ddot{\theta}_1$$

$$J_{O_1} \ddot{\theta}_1 = \tau_{r_1} - D_1 \dot{\theta}_1 - f_c r_1$$

$$\tau_{r_1} = \longrightarrow \textcircled{1}$$

$$\Sigma M_{O_1} = J_{O_1} \ddot{\theta}_1$$

$$J_{O_1} \ddot{\theta}_1 = \tau_{r_1} - f_c r_1 - D_1 \dot{\theta}_1$$

$$\tau_{r_1} = \longrightarrow \textcircled{1}$$

$$\Sigma M_{O_2} = -J_{O_2} \ddot{\theta}_2$$

$$-J_{O_2} \ddot{\theta}_2 = D_2 \dot{\theta}_1 - f_c r_1 - \tau_{r_2}$$

$$f_c = \left(J_{O_2} \ddot{\theta}_2 + D_2 \dot{\theta}_2 - \tau_{r_2} \right) \frac{1}{r_2} \longrightarrow \textcircled{2}$$

Rack & Pinion :-

$$\Sigma M_{O_3} = -J_{O_3} \ddot{\theta}_2$$

$$-J_{O_3} \ddot{\theta}_2 = \tau_{r_2} + D_3 \dot{\theta}_2 + f_c r_3$$

$$\tau_{r_2} = \longrightarrow \textcircled{3}$$

$$\Sigma M_{O_3} = -J_{O_3} \ddot{\theta}_2$$

$$-J_{O_3} \ddot{\theta}_2 = \tau_{R_2} + D_3 \dot{\theta}_2 + f_{C_2} r_3$$

$$\tau_{R_2} = \longrightarrow \textcircled{4}$$

$$\Sigma F_x = -m_R \ddot{x}$$

$$-m_R \ddot{x} = D_R \dot{x} - f_{C_2}$$

$$f_{C_2} = m_R \ddot{x} + D_R \dot{x}$$

$$\Sigma F_x = -m_R \ddot{x}$$

$$-m_R \ddot{x} = D_R \dot{x} - f_{C_2}$$

PD controller,

$$\bar{J} \ddot{\theta} + \bar{D} \dot{\theta} = v(t)$$

$$v(t) = \bar{J} \left[-K_P (\theta_m - \theta_d) - K_D (\dot{\theta}_m - \dot{\theta}_d) - K_I \int_0^t (\theta_m - \theta_d(\tau)) d\tau \right] + \bar{D} \dot{\theta} - \bar{J} \ddot{\theta}_d$$

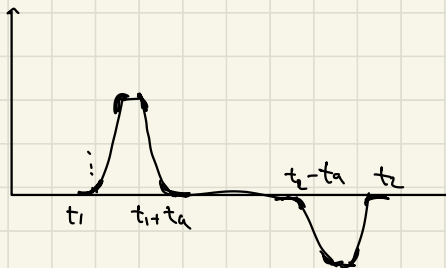
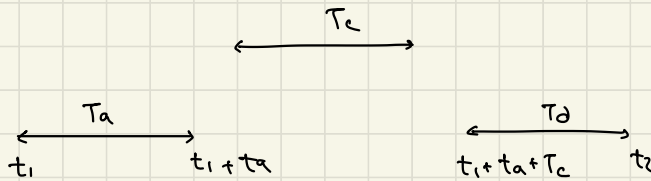
$$\bar{J} \ddot{\theta} + \bar{D} \dot{\theta} = v(t)$$

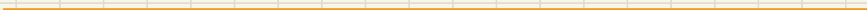
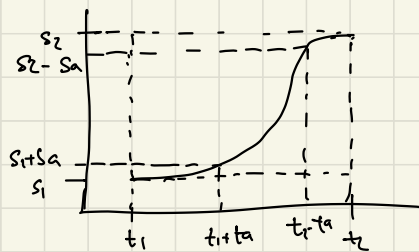
$$v(t) = \bar{J} \left[-k_p (\theta_m - \theta_{md}) - k_D (\dot{\theta}_m - \dot{\theta}_{md}) - k_I \int_0^t (\theta_m - \theta_{md}) d\tau \right] + \bar{D} \dot{\theta} + \bar{J} \ddot{\theta}$$

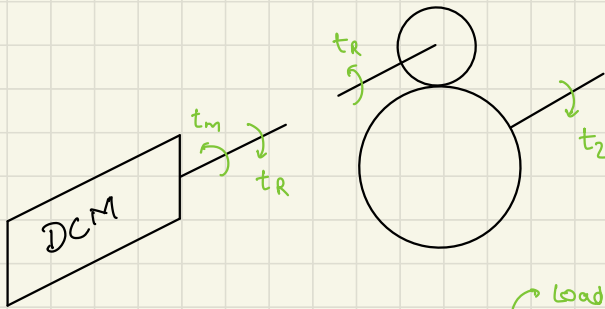
$$\bar{J} [(\ddot{\theta} - \ddot{\theta}_d) + k_D (\dot{\theta}_m - \dot{\theta}_{md}) + k_p (\theta - \theta_d)] = 0$$

$$\ddot{e} + k_D \dot{e} + k_p e + k_I e = 0$$

$$\lambda^3 + k_D \lambda^2 + k_p \lambda + k_I = (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$$







$$\frac{t_R}{\pi_1} = \frac{t_L}{\pi_2} \quad \omega$$

$$t_R = \frac{\pi_1}{\pi_2} t_L$$

$$\{\text{Motor Dynamics}\} = t_m(t) - t_R(t)$$

$$\ddot{\theta}_m, \dot{\theta}_m = t_m(t) - \frac{\pi_1}{\pi_2} t_2(t)$$

ACD = TVP
 Acceleration Cruise Deceleration

TVP :-

$$\left[s_1, s_2 \right] \quad \text{in} \quad \left[t_1, t_2 \right]$$

S T

$$v_{\max} = \frac{S_c}{T_c}$$

$$S_a = \frac{1}{2} a T_a^2 \quad \therefore \quad a_{\max} = \frac{2 S_a}{T_a^2}$$

$$\frac{2 S_a}{T_a} = \frac{S_c}{T_c}$$

In exam, assume S_a , if Q asks a_{max} below certain level, then reiterate.

$A_{max} \rightarrow$ low
 $S_a \rightarrow$ higher

max acceleration
 $\hookrightarrow a = \frac{2 S_a}{T_a^2}$

$\downarrow v_c = a T_a$
 constant velocity

LAB

$$\begin{array}{rcl} \text{Right} & = & 1280 \quad / \quad 1536 \\ \text{Left} & = & 2560 \quad / \quad 2816 \end{array} \quad \Bigg|$$

$$\text{fluctuation} = 256 \quad / \quad 256$$

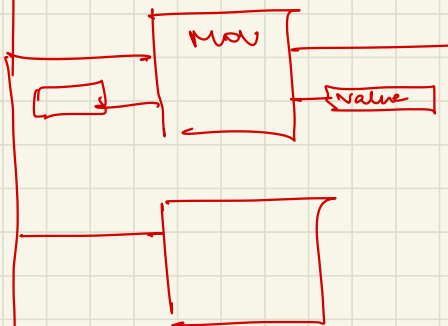
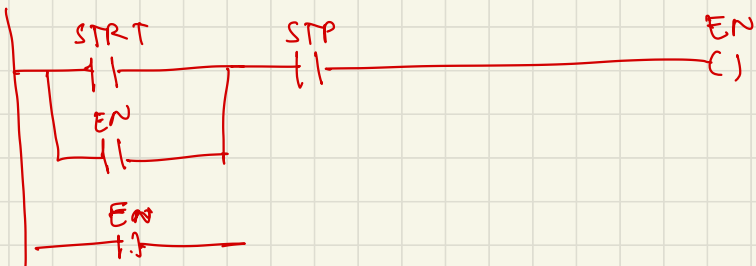
$$\text{mid - point} = 1920 \quad / \quad 2176$$

MR - Move Right
ML - Move Left

if (setp > Right val)
Move Left

setp > 1536
Mov. ~~setp~~

else if (setp < Left val)
Move Right



When not neglecting $L_a \frac{d}{dt} i_a$

$$L_a \frac{d}{dt} i_a + R_a i_a + K_b \dot{\theta}_m = v(t)$$

①

Differentiate

$$\bar{J} \ddot{\theta}_m + \bar{D} \dot{\theta}_m = t_m(t)$$

$$\bar{J} \ddot{\theta}_m + \bar{D} \dot{\theta}_m = \dot{t}_m(t)$$

$$t_m = K_T i_a$$

$$\frac{t_m}{K_T} = i_a$$

Substitute in ①,

$$L_a \frac{d}{dt} (i_a) + R_a i_a + K_b \dot{\theta}_m = v(t)$$

$$L_a \frac{\dot{t}_m}{K_T} + R_a i_a + K_b \dot{\theta}_m = v(t)$$

Convert to State Space Form for Simulation

Method I :- Differentiation

$$\begin{aligned} \bar{J} \ddot{\theta}_m + \bar{D} \dot{\theta}_m &= v(t) \\ \bar{J} \ddot{\theta}_m + \bar{D} \dot{\theta}_m &= \dot{v}(t) \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} v(t) = \bar{J} & \left[-K_p (\theta_m - \theta_d) - K_d (\dot{\theta}_m - \dot{\theta}_d) \right. \\ & \left. - K_I \int_0^t (\theta_m(\tau) - \theta_d(\tau)) d\tau \right] \\ & + \bar{J} \ddot{\theta}_d + \bar{D} \dot{\theta}_m \end{aligned}$$

Differentiating $v(t)$ & putting in (1)

$$\begin{aligned} \bar{J} \ddot{\theta}_m + \bar{D} \dot{\theta}_m &= \bar{J} \left(-K_p (\dot{\theta}_m - \dot{\theta}_d) - K_d (\ddot{\theta}_m - \ddot{\theta}_d) \right. \\ & \left. - K_I (\theta_m - \theta_d) \right) + \bar{J} \ddot{\theta}_d + \bar{D} \dot{\theta}_m \end{aligned}$$

error equation :-

$$\bar{J} (\ddot{e} + k_d \dot{e} + k_p e + k_I e) = 0$$

For state space form :-

$$\begin{aligned} \theta_1 &= \theta_1 & \dot{\theta}_1 &= \theta_2 \\ \dot{\theta}_m &= \theta_2 & \dot{\theta}_2 &= \theta_3 \\ \dot{\theta}_m &= \theta_3 \end{aligned}$$

$$\begin{aligned}
 \bar{J} \ddot{\theta}_3 &= \bar{J} \ddot{\theta}_m = -\bar{D} \dot{\theta}_1 + \bar{D} \dot{\theta}_1 \\
 &+ \bar{J} \left(-K_p (\theta_1 - \dot{\theta}_d) \right. \\
 &\quad \left. - K_D (\dot{\theta}_1 - \ddot{\theta}_d) \right. \\
 &\quad \left. - K_I (\theta_1 - \theta_d) \right) + \bar{J} \ddot{\theta}_d
 \end{aligned}$$

Method II :- Without Differentiation

$$\bar{J} \ddot{\theta}_m + \bar{D} \dot{\theta}_m = r(t)$$

$$\begin{aligned}
 r(t) &= \bar{J} \left[-K_p (\theta_m - \theta_d) - K_D (\dot{\theta}_m - \dot{\theta}_d) \right. \\
 &\quad \left. - K_I \int_0^t (\theta_m(\tau) - \theta_d(\tau)) d\tau \right] \\
 &+ \bar{J} \ddot{\theta}_d + \bar{D} \dot{\theta}_m
 \end{aligned}$$

$$\theta_1 = \int_0^t \theta_m(\tau) d\tau \quad \dot{\theta}_1 = \theta_2$$

$$\theta_2 = \dot{\theta}_m(t) \quad \dot{\theta}_2 = \theta_3$$

$$\theta_3 = \ddot{\theta}_m$$

$$\begin{aligned}
 \bar{J} \ddot{\theta}_3 &= \bar{J} \ddot{\theta}_m = \bar{J} \left[-K_p (\theta_2 - \dot{\theta}_d) - K_D (\theta_3 - \ddot{\theta}_d) \right. \\
 &\quad \left. - K_I \left(\theta_1 - \int_0^t \theta_d(\tau) d\tau \right) \right] \\
 &+ \bar{J} \ddot{\theta}_d + \bar{D} \theta_3
 \end{aligned}$$

SSSS abseck

SSS abseleation

SS absecity

S absement

displacement

• velocity

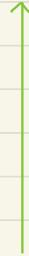
•• acceleration

••• jerk

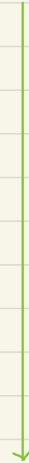
•••• jounce

••••• founce

•••••• bounce

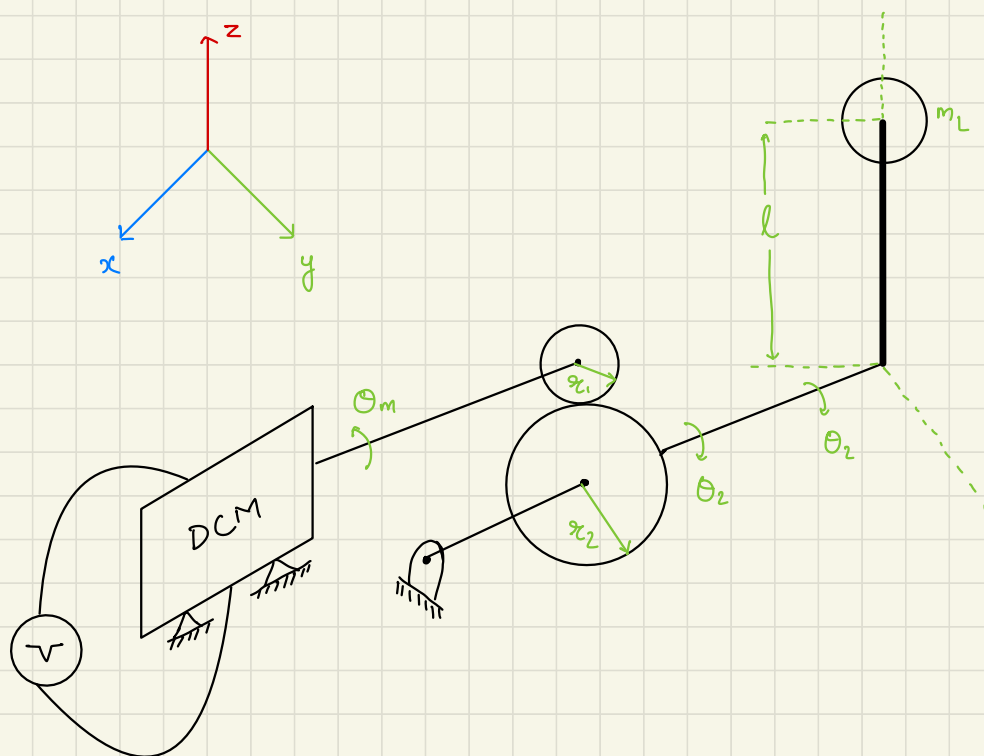


Integration

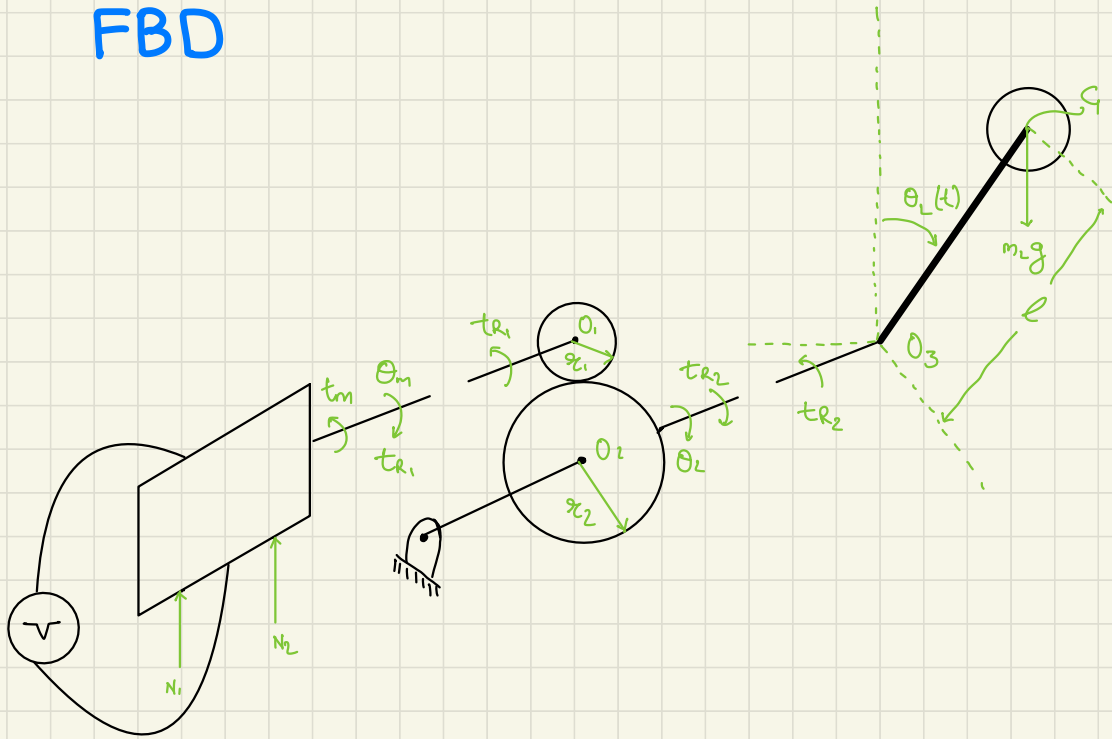


Differentiation

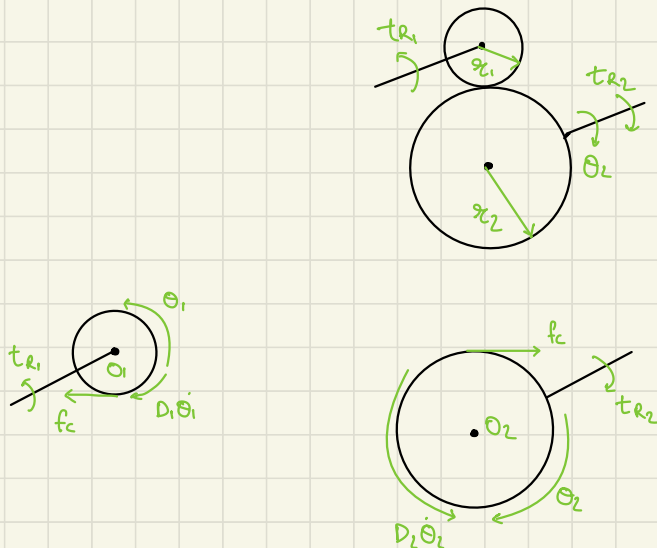
Position Control of Inverted Pendulum



FBD



Gearbox :-



Motor :-

(Electrical
eqⁿ)

$$v(t) = L_a \frac{di_a}{dt} + R_a i_a + K_b \dot{\theta}_m$$

$$t_m = K_T i_a$$

$$\frac{R_a t_m}{K_T} + K_b \dot{\theta}_m = v(t) \quad \text{---} \quad (1)$$

(Mechanical
eqⁿ)

$$J_m \ddot{\theta}_i + D_m \dot{\theta}_i = t_m$$

$$\frac{R_a}{K_T} (J_m \ddot{\theta}_i + D_m \dot{\theta}_i) + K_b \dot{\theta}_m = v(t)$$

$$J_m \ddot{\theta}_i + D_m \dot{\theta}_i + \frac{K_b K_T}{R_a} \dot{\theta}_i = K_b K_T v(t)$$

$$J_m \ddot{\theta}_i + D_m \dot{\theta}_i = t_m(t) - t_{r,i}(t) \quad \text{---} \quad (2)$$

Gearbox :-

$$\Sigma M_{O_1} = J_{O_1} \ddot{\theta}_i$$

$$J_{O_1} \ddot{\theta}_i = t_{r,i} - f_{c,i} r_i - D_i \dot{\theta}_i$$

$$t_{r,i} = J_{O_1} \ddot{\theta}_i + f_{c,i} r_i + D_i \dot{\theta}_i$$

$$\text{---} \quad (3)$$

$$\Sigma M_{O_2} = -J_{O_2} \ddot{\theta}_L$$

$$-J_{O_2} \ddot{\theta}_L = D_2 \dot{\theta}_L - f_{c1} r_2 - \tau_{R_2}$$

$$f_{c1} r_2 = J_{O_2} \ddot{\theta}_L + D_2 \dot{\theta}_L - \tau_{R_2}$$

$$f_{c1} = (J_{O_2} \ddot{\theta}_L + D_2 \dot{\theta}_L - \tau_{R_2}) \frac{1}{r_2}$$

—————→ (4)

Inverted pendulum :-

$$\Sigma M_{O_3} = -J_{O_3} \ddot{\theta}_L$$

$$-J_{O_3} \ddot{\theta}_L = D_2 \dot{\theta}_L + \tau_{R_2} - m_2 g \sin \theta_L$$

$$-\tau_{R_2} = D_2 \dot{\theta}_L + J_{O_3} \ddot{\theta}_L - m_2 g \sin \theta_L$$

—————→ (5)

EOM :-

$$f_{c1} = \overset{(5)}{\text{in}} \overset{(4)}{\text{,}} (J_{O_2} \ddot{\theta}_L + D_2 \dot{\theta}_L + D_2 \dot{\theta}_L + J_{O_3} \ddot{\theta}_L$$

$$- m_2 g \sin \theta_L) \frac{1}{r_2}$$

—————→ (6)

⑥ in ③

$$t_{R_1} = J_{O_1} \ddot{\theta}_1 + \frac{x_1}{x_2} \left[J_{O_2} \ddot{\theta}_L + 2D_2 \dot{\theta}_L + J_{O_3} \ddot{\theta}_L - m_L g \sin \theta_L \right] + D_1 \dot{\theta}_1$$

⑦

Put ⑦ in ②

$$J_m \ddot{\theta}_i + D_m \dot{\theta}_i = \tau_m(t) - J_{o1} \ddot{\theta}_i - \frac{\tau_1}{\tau_2} \left[J_{o2} \ddot{\theta}_i + 2D_2 \dot{\theta}_i + J_{o3} \ddot{\theta}_i - m_L g \sin \theta_L \right] - D_1 \dot{\theta}_i$$

$$t_m(t) = -\ddot{\Theta}_1 [J_m + J_{O_1}] - \ddot{\Theta}_1 [D_m + D_1] - \frac{\pi_1}{\pi_2} \left[J_{O_2} \ddot{\Theta}_1 + 2D_2 \dot{\Theta}_1 + J_{O_3} \ddot{\Theta}_1 - m_L g \sin \Theta_L \right]$$

check the signs

1 → 8

Put ⑧ in ①

$$\frac{R_a}{K_T} \left[-\ddot{\theta}_1 [J_m + J_{o1}] - \dot{\theta}_1 [D_m + D_1] - \frac{r_1}{r_2} \left[J_{o2} \ddot{\theta}_1 + 2D_2 \dot{\theta}_1 + J_{o3} \ddot{\theta}_1 - m_L g \sin \theta_L \right] + K_b \dot{\theta}_m = v(t) \right]$$

$$\begin{aligned} \theta_1 &= \theta_m \\ \theta_2 &= \theta_L \\ \theta_L &= \frac{r_1}{r_2} \theta_m \end{aligned}$$

use this

for → ⑧

EOM for inverted pendulum

$$\begin{aligned} \bar{J}\ddot{\theta}_m + \bar{D}\dot{\theta}_m + \left(\frac{\pi_1}{\pi_2}\right)' \left[J_{02}\ddot{\theta}_m + 2D_2\dot{\theta}_m + J_{03}\ddot{\theta}_m \right] \\ - \frac{\pi_1}{\pi_2} m_2 g l \sin\left(\frac{\pi_1}{\pi_2} \theta_m\right) = t_m(t) = K_T i_a \end{aligned}$$

Electrical eqⁿ of DCM :

$$L_a \frac{di_a}{dt} + R_a i_a + K_b \dot{\theta}_m = v(t)$$

$$t_m(t) = K_T i_a$$

$$i_a = \frac{t_m}{K_T}$$

$$\begin{aligned} \frac{R_a}{K_T} \left[\bar{J}\ddot{\theta}_m + \bar{D}\dot{\theta}_m + \left(\frac{\pi_1}{\pi_2}\right)' \left[\right. \right. \\ \left. \left. - \frac{\pi_1}{\pi_2} m_2 g l \sin\left(\frac{\pi_1}{\pi_2} \theta_m\right) \right] \right. \\ \left. + K_b \dot{\theta}_m \right] = v(t) \end{aligned}$$

Just combine J terms

$$\downarrow$$
$$J_T = \bar{J} + \left(\frac{\pi_1}{\pi_2}\right)^2 (J_{02} + J_{03})$$

$$D_T = \bar{D} + \left(\frac{\pi_1}{\pi_2}\right)^2 (2D_2) + K_b$$

$$J_T \ddot{\theta}_m + D_T \dot{\theta}_m - \frac{x_1}{\pi_2} m_2 g l \sin\left(\frac{\pi_1}{\pi_2} \theta_m\right) = v(t)$$

Computed Torque Control :-

PID Controller

$$\begin{aligned} v(t) = & -\frac{\pi_1}{\pi_2} m_2 g l \sin\left(\frac{\pi_1}{\pi_2} \theta_m\right) + D_T \dot{\theta}_m \\ & + J_T \left[\ddot{\theta}_d - K_p (\theta_m - \theta_d) \right. \\ & \quad \left. - K_D (\dot{\theta}_m - \dot{\theta}_d) \right. \\ & \quad \left. - K_I \int_0^t [\theta_m(\tau) - \theta_d(\tau)] d\tau \right] \end{aligned}$$

Control Law

$\swarrow$ $J\ddot{\theta}_d$ (take common)

(J_T is factored out & equal to 0)

$$\begin{aligned} & J_T (\ddot{\theta}_m - \ddot{\theta}_d) + J_T (K_D (\dot{\theta}_m - \dot{\theta}_d)) \\ & + J_T (K_p (\theta_m - \theta_d)) \\ & + J_T \left(K_I \left(\int_0^t (\theta_m(\tau) - \theta_d(\tau)) d\tau \right) \right) = 0 \end{aligned}$$

$$\ddot{e} + K_D \dot{e} + K_P e + K_I \int e(\tau) d\tau = 0$$

differentiating again ,

$$\lambda^3 + K_D \lambda^2 + K_P \lambda + K_I = 0$$

LAB

A+B+B-A-

$$Q \times 0.0 = A$$

$$Q \times 0.3 = B$$

$$\% I \times 0.1 = A(\text{ext})$$

$$\% I \times 0.3 = A(\text{retracted})$$

$$\% I \times 0.2 = B(\text{ext})$$

$$\% I \times 0.4 = B(\text{retracted})$$

$$\% I \times 0.0 = \text{Start}$$

$$\% I \times 0.5 = \text{Stop}$$

Script :-

Simulation of closed loop system of inverted pendulum

$$J\ddot{\theta} + c\dot{\theta} + k\theta - 0.5\sin\theta = \tau(t)$$

$$\begin{aligned}\tau(t) = & -0.5\sin\theta + c\dot{\theta} + k\theta \\ & + [\ddot{\theta}_d - K_p(\theta - \theta_d) \\ & - K_d(\dot{\theta} - \dot{\theta}_d) \\ & - K_I \left(\int_0^t (\theta(\tau) - \theta_d(\tau)) d\tau \right)] J\end{aligned}$$

$$\theta_1 = y(1, 1) \quad \int_0^t \theta(\tau) d\tau = \theta_1$$

$$\theta_2 = y(2, 1) \quad \dot{\theta}_1 = \theta(t) = \theta_2$$

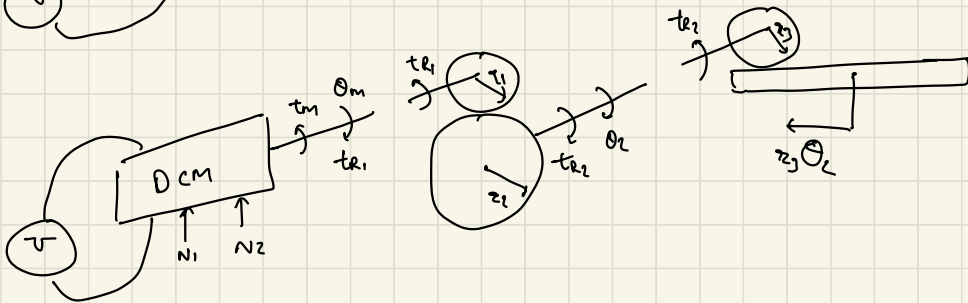
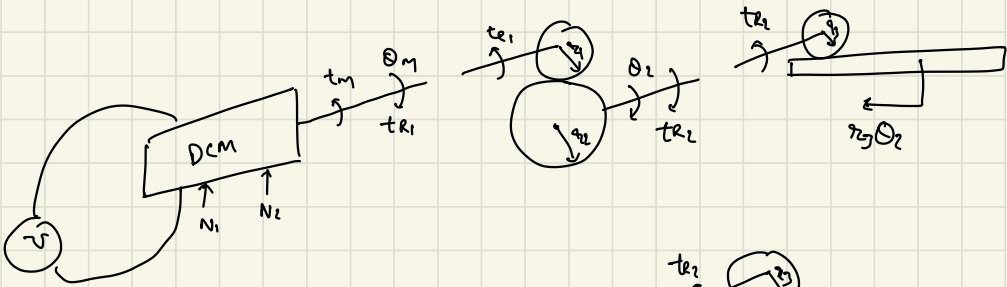
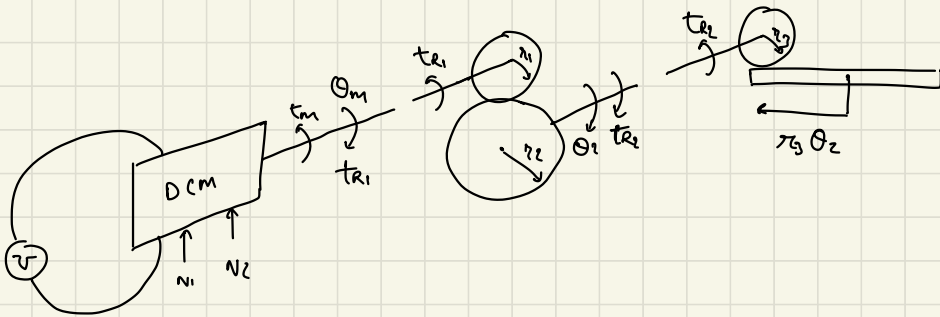
$$\theta_3 = y(3, 1) \quad \dot{\theta}_2 = \dot{\theta} = \theta_3$$

$$\begin{aligned}\dot{\theta}_3 = & 0.5\sin(y(2, 1)) + 0.5\sin(y(2, 1)) \\ & + (\theta_3 - \dot{\theta}_3 + k\theta_1 - k\theta_1 + [a \\ & - K_p(\theta_2 - s) - K_d(\theta_3 - v) - K_I(\theta_1 - s_i)]) J\end{aligned}$$

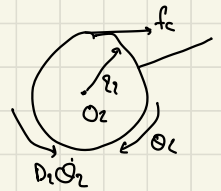
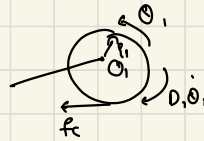
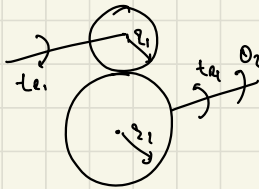
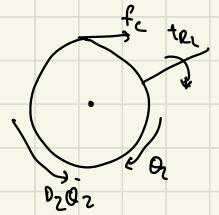
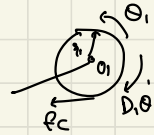
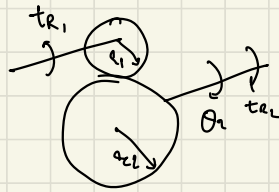
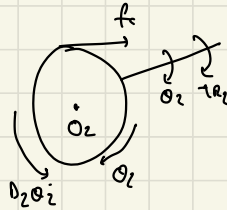
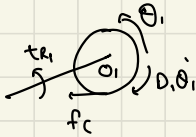
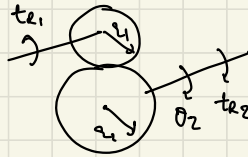
M-2

FBD

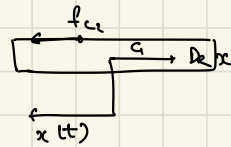
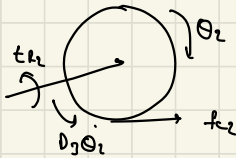
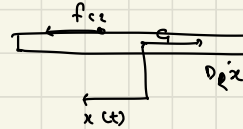
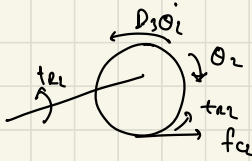
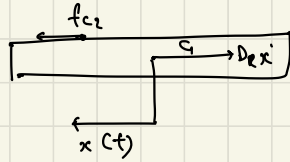
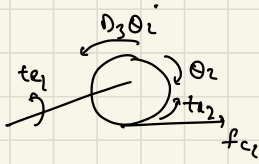
Rack & Pinion



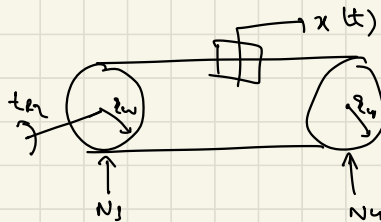
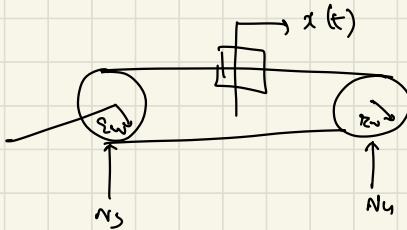
Gear box



Rack & Pinion



Belt drive



KINEMATICS

Rack & Pinion :-

$$\rightarrow \begin{aligned} \theta_1 r_1 &= \theta_2 r_2 \\ \theta_1 &= \theta_m \end{aligned}$$

$$\theta_m r_1 = \theta_2 r_2$$

$$x(t) = r_3 \theta_2(t)$$

$$\boxed{x(t) = r_3 \theta_m \frac{r_1}{r_2}}$$

$$\rightarrow \begin{aligned} \theta_1 r_1 &= \theta_2 r_2 \\ \theta_1 &= \theta_m \\ \theta_m r_1 &= \theta_2 r_2 \end{aligned}$$

$$x(t) = r_3 \theta_2(t)$$

$$\boxed{x(t) = r_3 \frac{r_1}{r_2} \theta_m(t)}$$

$$\rightarrow \begin{aligned} \theta_1 r_1 &= \theta_2 r_2 \\ \theta_1 &= \theta_m \\ \theta_m r_1 &= \theta_2 r_2 \end{aligned}$$

$$x(t) = r_3 \theta_2(t)$$

$$\boxed{x(t) = r_3 \theta_m \frac{r_1}{r_2}}$$

Belt Drive :- $\rightarrow \tau_1 \theta_1 = \tau_2 \theta_2$

$$\theta_1 = \theta_m$$

$$\theta_2 = \theta_w$$

$$\tau_1 \theta_m = \tau_2 \theta_w$$

$$x(t) = \tau_w \theta_w$$

$$x(t) = \tau_w \frac{\tau_1}{\tau_2} \theta_m(t)$$

$$\rightarrow \tau_1 \theta_1 = \tau_2 \theta_2$$

$$\theta_1 = \theta_m$$

$$\theta_2 = \theta_w$$

$$\tau_1 \theta_m = \tau_2 \theta_w$$

$$x(t) = \tau_w \theta_w$$

$$x(t) = \tau_w \frac{\tau_1}{\tau_2} \theta_m$$

$$\rightarrow \tau_1 \theta_1 = \tau_2 \theta_2$$

$$\theta_1 = \theta_m$$

$$\theta_2 = \theta_w$$

$$\tau_1 \theta_m = \tau_2 \theta_w$$

$$x(t) = \tau_w \theta_w$$

$$x(t) = \tau_w \frac{\tau_1}{\tau_2} \theta_m$$

EOM

Motor

$$v(t) = \cancel{L_a \frac{di_a}{dt}}^0 + R_a i_a + K_b \dot{\theta}_m$$

$$v(t) = R_a i_a + K_b \dot{\theta}_m$$

$$t_m = K_T i_a$$

$$R_a \frac{t_m}{K_T} + K_b \dot{\theta}_m = v(t)$$

→ ①

$$J_0 \ddot{\theta}_i + D_0 \dot{\theta}_i = t_m$$

$$\frac{R_a}{K_T} (J_0 \ddot{\theta}_i + D_0 \dot{\theta}_i) + K_b \dot{\theta}_m = v(t)$$

$$J_0 \ddot{\theta}_i + D_0 \dot{\theta}_i = t_m(t) - t_{e1}(t)$$

→ ②

Motor

$$v(t) = \cancel{L_a \frac{di_a}{dt}}^0 + R_a i_a + K_b \dot{\theta}_m$$

$$R_a i_a + K_b \dot{\theta}_m = v(t)$$

$$t_m = K_T i_a$$

$$R_a \frac{t_m}{K_T} + K_b \dot{\theta}_m = v(t)$$

→ ①

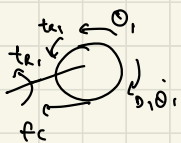
$$J_0 \ddot{\theta}_i + D_0 \dot{\theta}_i = t_m$$

$$\frac{R_a}{K_T} (J_0 \ddot{\theta}_i + D_0 \dot{\theta}_i) + K_b \dot{\theta}_m = v(t)$$

$$J_0 \ddot{\theta}_i + D_0 \dot{\theta}_i = t_m(t) - t_{e1}$$

→ ②

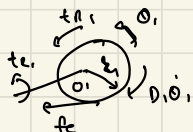
Gear box



$$\Sigma M_{O1} = J_{O1} \ddot{\theta}_1$$

$$J_{O1} \ddot{\theta}_1 = tr_1 - f_{c1} r_1 - D_1 \dot{\theta}_1$$

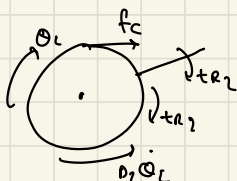
$$tr_1 = \longrightarrow \textcircled{3}$$



$$\Sigma M_{O2} = J_{O2} \ddot{\theta}_2$$

$$J_{O2} \ddot{\theta}_2 = tr_2 - f_{c2} r_2 - D_2 \dot{\theta}_2$$

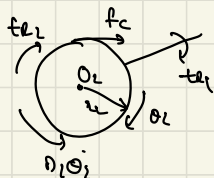
$$tr_2 = \longrightarrow \textcircled{4}$$



$$\Sigma M_{O3} = -J_{O3} \ddot{\theta}_3$$

$$-J_{O3} \ddot{\theta}_3 = D_3 \dot{\theta}_3 - f_{c3} r_3 - tr_3$$

$$f_{c1} = \longrightarrow \textcircled{4}$$

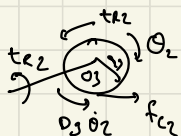


$$\Sigma M_{O4} = -J_{O4} \ddot{\theta}_4$$

$$-J_{O4} \ddot{\theta}_4 = D_4 \dot{\theta}_4 - f_{c4} r_4 - tr_4$$

$$f_{c1} = \longrightarrow \textcircled{4}$$

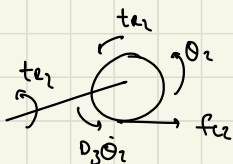
Rack & Pinion



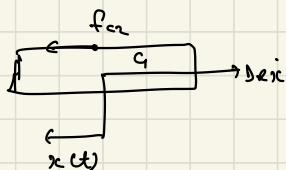
$$\Sigma M_{O3} = -J_{O3} \ddot{\theta}_3$$

$$-J_{O3} \ddot{\theta}_3 = r_3 f_{c1} + tr_2 + D_3 \dot{\theta}_3$$

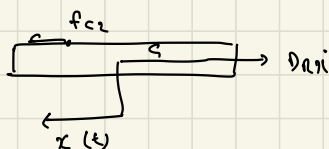
$$tr_2 = \longrightarrow \textcircled{5}$$



$$\begin{aligned}\Sigma M_{O_3} &= -J_{O_3} \ddot{\theta}_1 \\ -J_{O_3} \ddot{\theta}_1 &= D_3 \dot{\theta}_1 + f_{c2} r_3 + t_{\theta_1} \\ t_{\theta_1} &= \longrightarrow \textcircled{5}\end{aligned}$$



$$\begin{aligned}\Sigma F_x &= -m_e \ddot{x} \\ -m_e \ddot{x} &= -f_{c2} + D_{ex} \dot{x} \\ f_{c2} &= \longrightarrow \textcircled{6}\end{aligned}$$



$$\begin{aligned}\Sigma F_x &= -m_e \ddot{x} \\ -m_e \ddot{x} &= -f_{c2} + D_{ex} \dot{x} \\ f_{c2} &= \longrightarrow \textcircled{6}\end{aligned}$$

COMPUTED TORQUE CONTROL

Belt Drive

$$\bar{J} \ddot{\theta} + \bar{D} \dot{\theta} = v(t)$$

$$\begin{aligned}v(t) = \bar{J} \left[-k_p (\theta - \theta_d) \right. \\ \left. - k_v (\dot{\theta} - \dot{\theta}_d) \right. \\ \left. - k_I \int_0^t [\theta(\tau) - \theta_d(\tau)] d\tau \right] \\ + \bar{D} \dot{\theta} + \bar{J} \ddot{\theta}_d\end{aligned}$$

$$\bar{J} \ddot{\theta} + \bar{D} \dot{\theta} = v(t)$$

$$\begin{aligned}v(t) = \bar{J} \left[-k_p (\theta - \theta_d) - k_v (\dot{\theta} - \dot{\theta}_d) \right. \\ \left. - k_I \int_0^t [\theta(\tau) - \theta_d(\tau)] d\tau \right] \\ + \bar{D} \dot{\theta} + \bar{J} \ddot{\theta}_d\end{aligned}$$

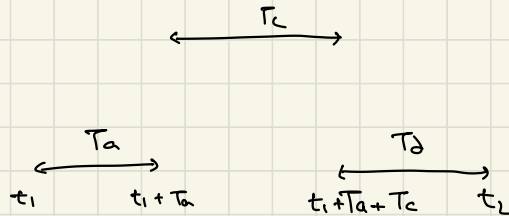
$$\int_0^T [(\ddot{O} - \ddot{O}_d) + K_D (\dot{O} - \dot{O}_d) + K_P (O - O_d) + K_I \int_0^t [O(\tau) - O_d(\tau)] d\tau] d\tau = 0$$

differentiate

$$\ddot{e} + K_D \dot{e} + K_P e + K_I \int e = 0$$

$$\lambda^3 + K_D \lambda^2 + K_P \lambda + K_I = (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$$

for T_{vp}

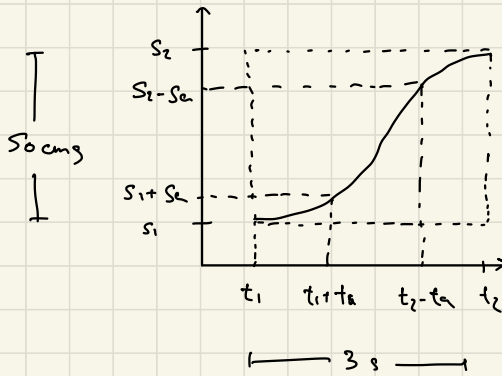


Assume $S_R = S_{cr}$
 $S_a = S_d = S_{cr}$

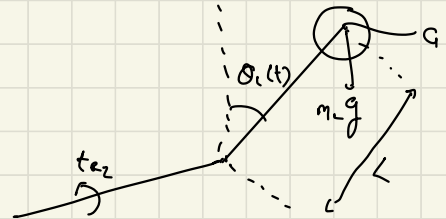
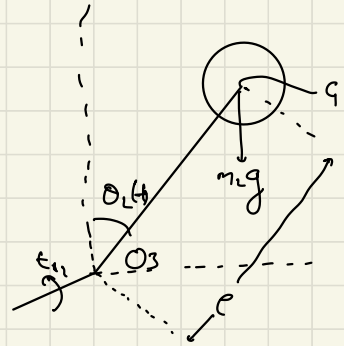
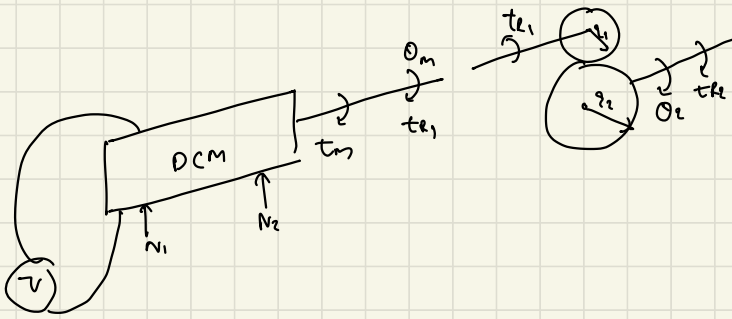
$$S_c = S - S_a - S_d$$

$$T_a = \frac{S_a T}{2S_a + \frac{S_c}{2}}$$

$$T_a = \frac{S_a T}{2S_a + \frac{S_c}{2}}$$



FBD inv pend



EOM for inv pend

$$\begin{aligned}\Sigma M_{O_3} &= -J_{O_3} \ddot{\theta}_l \\ -J_{O_3} \ddot{\theta}_l &= \tau_{e2} + D_2 \dot{\theta}_l - m_2 g \sin \theta_l \\ \tau_{e2} &= \longrightarrow \textcircled{5}\end{aligned}$$

$$\begin{aligned}\Sigma M_{O_3} &= -J_{O_3} \ddot{\theta}_1 \\ -J_{O_3} \ddot{\theta}_1 &= \tau_{\tau_2} + D_2 \dot{\theta}_1 - m_2 g \sin \theta_1 \\ \tau_{\tau_2} &= \longrightarrow \textcircled{S}\end{aligned}$$

$$\begin{aligned}\Sigma M_{O_3} &= -J_{O_3} \ddot{\theta}_1 \\ -J_{O_3} \ddot{\theta}_1 &= \tau_{\tau_2} + D_2 \dot{\theta}_1 - m_2 g \sin \theta_1 \\ \tau_{\tau_2} &= \longrightarrow \textcircled{S}\end{aligned}$$

Computed Torque Control

$$\ddot{\theta}_d + \ddot{\theta}_0 = v(t)$$

$$v(t) = -\frac{x_1}{x_2} m_2 g \sin\left(\frac{x_1}{x_2} \theta_m\right) + D_T \dot{\theta}_m$$

$$\begin{aligned}+ J_T \left[\ddot{\theta}_d - K_P (\theta_m - \theta_d) \right. \\ \left. - K_D (\dot{\theta}_m - \dot{\theta}_d) \right. \\ \left. - K_I \int_0^t [\theta_m(\tau) - \theta_d(\tau)] d\tau \right]\end{aligned}$$

$$\begin{aligned}v(t) = -\frac{x_1}{x_2} m_2 g \sin\left(\frac{x_1}{x_2} \theta_m\right) + D_T \dot{\theta}_m \\ + J_T \left[\ddot{\theta}_d - K_P (\theta_m - \theta_d) \right. \\ \left. - K_D (\dot{\theta}_m - \dot{\theta}_d) \right. \\ \left. - K_I \int_0^t [\theta_m(\tau) - \theta_d(\tau)] d\tau \right]\end{aligned}$$

$$\tau(t) = -\frac{r_1}{r_2} m_L g \sin\left(\frac{r_1}{r_2} \theta_m\right) + J_T (\ddot{\theta}_d - k_p \dots)$$

$$\begin{aligned} J_T (\ddot{\theta}_m - \ddot{\theta}_d) + J_T (k_p (\theta_m - \theta_d)) \\ + J_T (k_v (\dot{\theta}_m - \dot{\theta}_d)) \\ + J_T \left(k_I \int_0^t (\theta_m(\tau) - \theta_d(\tau)) d\tau \right) = 0 \end{aligned}$$

$$\ddot{\theta} + k_D \dot{\theta} + k_P \theta + k_I \int \theta = 0$$

$$E(\text{var. rot}) = I \frac{d\dot{\alpha}}{dt} + R \dot{\alpha} \dot{\alpha} + K_b \dot{\alpha} \rightarrow \frac{R \dot{\alpha}^2}{K_f} + K_b \dot{\alpha}$$

$$\text{Mech: } J \ddot{\alpha} + D \dot{\alpha} = b \alpha(t)$$

$$\frac{R_c (J \ddot{\alpha} + D \dot{\alpha})}{K_f} + K_b \dot{\alpha}$$

$$\Sigma M_{O_1} = T \cdot \ddot{\alpha}_1$$

$$= R_1 - f R_1 - D \dot{\alpha}_1$$

$$\Sigma M_{O_2} = \int \ddot{\alpha}_2$$

$$= D \dot{\alpha}_2 - f e_1 - t e_2$$

$$\theta_{11} = \theta_{21} \theta_2$$

$$\dot{\theta} = \frac{1}{2} \dot{\theta}_2$$

$$f_c:$$

$$f_c \ddot{\alpha}$$

$$r(\alpha) = \int [-K_b(\theta - \dot{\alpha}) - K_b(\ddot{\alpha} - \ddot{\alpha}) - K_b \ddot{\alpha} + \ddot{\alpha} \ddot{\alpha}] + \ddot{\alpha} + \ddot{\alpha} \ddot{\alpha} \quad 40 + 50$$

$$\ddot{\alpha} [\ddot{\alpha} - \ddot{\alpha}] - K_b \ddot{\alpha} - K_f (\theta - \dot{\alpha}) = 0$$

$$\ddot{\alpha} + K_b \ddot{\alpha} + K_f \theta = 0$$

$$\lambda^2 + (K_b \lambda + K_f) = 0 \quad (\lambda + \lambda_1)(\lambda + \lambda_2)$$

$$\lambda^2 + K_b \lambda + K_f = (\lambda + 40)(\lambda + 50) = 0$$

$$= \lambda(\lambda + 40) + 50(\lambda + 40) = \lambda^2 + 90\lambda + 2000$$

$$r(t) = \ddot{\alpha}$$

$$\ddot{\alpha} [-K_b$$

$$\ddot{\alpha} = \ddot{\alpha}$$

$$\ddot{\alpha} = \ddot{\alpha}$$

$$\ddot{\alpha} + K_b \ddot{\alpha} + K_f \theta = 0$$

$$\lambda^2 + K_b \lambda + K_f = 0$$

$$K_b = 90$$

$$K_f = 2000$$

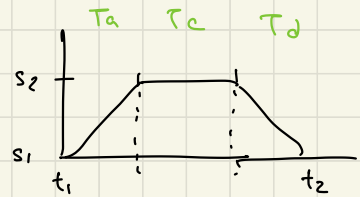
$$\lambda^2 + K_b \lambda + K_f = (\lambda + 40)(\lambda + 50) = 0$$

Function of Time TVP

$$a = \frac{2s_a}{T_a^2}$$

when $t \in [t_1, t_1 + T_a]$

$$x(t) = s_1 + \frac{1}{2} a t_3^2 \quad \rightarrow t - t_1$$



when $t \in [t_1 + T_a, t_1 + T_a + T_c]$

$$x(t) = s_1 + s_c + v_c t_4 \quad \rightarrow t_1 - (t_1 + T_a)$$

when $t \in [t_1 + T_a + T_c, t_1 + T]$

$$x(t) = s_1 + s_c + s_a + v_c t_6 - \frac{1}{2} a t_6^2 \quad \rightarrow t - [t_1 + T_a + T_c]$$

$$a = \frac{2s_a}{T_a^2}$$

when $T \in [t_1, t_1 + T_a]$

$$x(t) = s_1 + \frac{1}{2} a t_3^2 \quad \rightarrow t - t_1$$

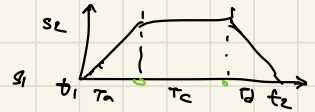
when $t \in [t_1 + T_a, t_1 + T_a + T_c]$

$$x(t) = s_1 + s_a + v t_4 \quad \rightarrow t - [t_1 + T_a]$$

when $t \in [t_1 + T_a + T_c, t_1 + T]$

$$x(t) = s_1 + s_a + s_c + v_c t_6 - \frac{1}{2} a t_6^2$$

when $T \in [t_1, t_1 + T_a]$,



$$x(t) = s_1 + \frac{1}{2} a t_3^2$$

$\xrightarrow{\quad} t - t_1$

when $T \in [t_1 + T_a, t_1 + T_a + T_c]$,

$$x(t) = s_1 + s_a + v_c t_4$$

$\xrightarrow{\quad} t - (t_1 + T_a)$

when $T \in [t_1 + T_a + T_c, t_1 + T]$

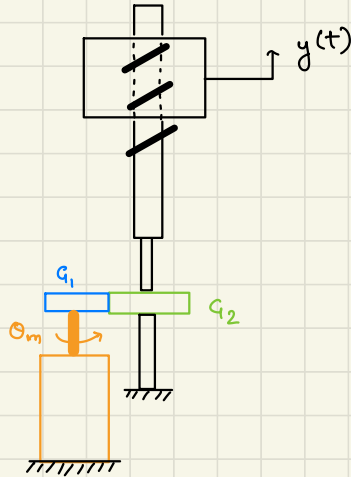
$$x(t) = s_1 + s_a + s_c + v_c t_6 - \frac{1}{2} a t_6^2$$

$\downarrow$
 $t - [t_1 + T_a + T_c]$

POWER SCREW

(simplified)

With Gearbox :-



Hw 2

$$P = (0.3, 0.5)$$

$$Q = (0.7, -0.1)$$

$$O = (0, 0) \quad \text{origin}$$

$$S_1 = \sqrt{0.3^2 + 0.5^2} = 0.58$$

distance b/w PQ :

$$S = \sqrt{0.4^2 + 0.5^2} = 0.7211$$

$$t_1 = 3s$$

$$t_0 = 2s$$

$$T = S_s$$

$$a_{\max} = \frac{2S_1}{t_1^2} = \frac{2(0.58)}{9}$$

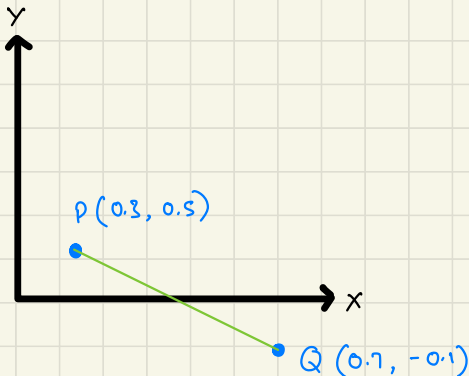
$$a_{\max} = 0.128 \text{ m/s}^2$$

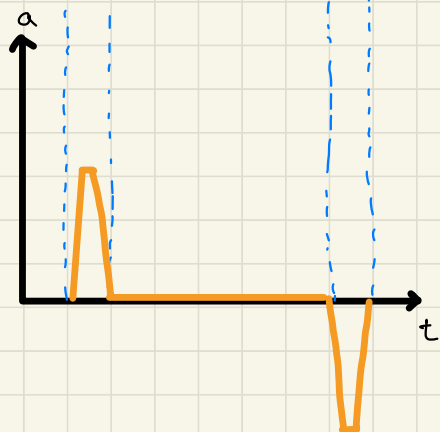
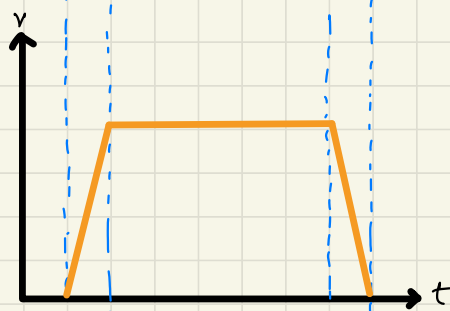
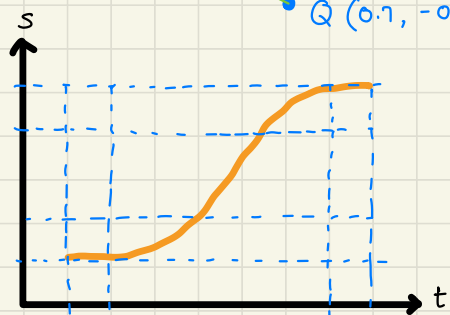
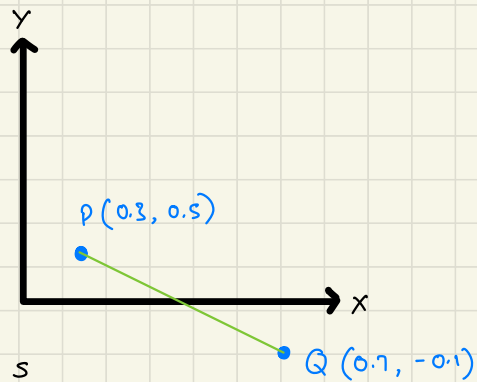
which is under the given
 $a_{\max}$ constraint (1 m/s^2)

$$S_a = 0.025$$

$$S_{aa} = 0.005$$

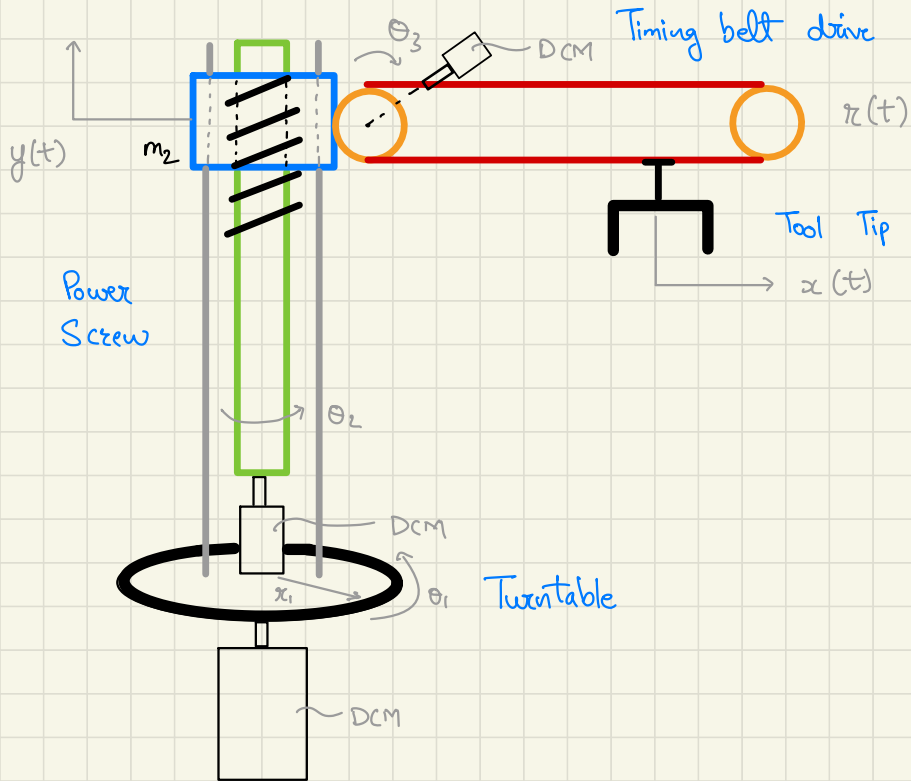
$$t = 0.10$$





HW 3

Q. Turntable + Power Screw + Timing Belt



Here , $x(t) = \alpha_1(t) \cdot \theta_3$

$$y(t) = \frac{\theta_2}{2\pi} \lambda \rightarrow \text{Lead}$$

3 motors to power each component respectively.

No. of EOMs = 3

$$\text{Tooltip} \equiv (\theta_1, y(t), x(t))$$

1. Turntable :- $\text{DCM} :- \frac{R_a}{K_T} t_{m_1} + K_b \dot{\theta}_m = v_r(t)$

$$J_m \ddot{\theta}_1 + D_m \dot{\theta}_1 = t_{m_1}(t) - t_{r_1}(t)$$

$$\text{GB} :- \sum M_{O_1} = J_{O_1} \ddot{\theta}_1$$

$$J_{O_1} \ddot{\theta}_1 = t_{r_1} - f_{c,r_1} - D_1 \dot{\theta}_1$$

$$t_{r_1} = J_{O_1} \ddot{\theta}_1 + f_{c,r_1} + D_1 \dot{\theta}_1$$

$$J_m \ddot{\theta}_1 + D_m \dot{\theta}_1 = t_{m_1}(t) - J_{O_1} \ddot{\theta}_1 - f_{c,r_1} - D_1 \dot{\theta}_1$$

$$2 J_m \ddot{\theta}_1 + 2 D_m \dot{\theta}_1 + f_{c,r_1} = t_{m_1}(t)$$

$\therefore \text{EOM}_1 \Rightarrow$

$$\boxed{\frac{R_a}{K_T} (2 J_m \ddot{\theta}_1 + 2 D_m \dot{\theta}_1 + f_{c,r_1}) + K_b \dot{\theta}_m = v_r(t)}$$

2. Power Screw :-

$$DCM :- \frac{R_a}{K_r} \dot{m}_2 + K_b \frac{2\pi}{\lambda} \dot{y} = \tau(t)$$

$$J_{m_2} \ddot{\theta}_2 + D_{m_2} \dot{\theta}_2 = \dot{m}_2(t) - \dot{r}_2(t)$$

$$\text{Slide :- } \Sigma F_y = m \ddot{y} = f - m g$$

$$f = m_2 \ddot{y} + m_2 g$$

$$G(u, \alpha) = \frac{y \cos \alpha + x \sin \alpha}{\cos \alpha - y \sin \alpha}$$

$$\text{Screw :- } J_{sc} \frac{2\pi}{\lambda} \ddot{y} + G(u, \alpha) r_{sc} m \ddot{y} + G(u, \alpha) r_{sc} m g = \dot{r}(t)$$

$$\therefore \text{Eom}_2 \Rightarrow \left(J_{sc} + J_{m_2} \right) \frac{2\pi}{\lambda} \ddot{y} + G(u, \alpha) r_{sc} m \ddot{y} = \dot{m}_2(t) + D_{m_2} \frac{2\pi}{\lambda} \dot{y} + G(u, \alpha) r_{sc} m g$$

3. Timing belt Drive

$$\text{Motor :- } \frac{R_a \dot{m}_3}{K_r} + K_b \dot{\theta}_3 = \tau_3(t)$$

$$J_{m_3} \ddot{\theta}_3 + D_{m_3} \dot{\theta}_3 = \dot{m}_3(t) - \dot{r}_3(t)$$

Belt Drive :-

$$\begin{aligned} & - [J_{O_3} + J_{O_4} m r \omega^2] \ddot{\theta}_3 \\ & - [D_3 + D_4] \dot{\theta}_3 \end{aligned} = \tau_{e_3}(t)$$

So,

$$J_{m_3} \ddot{\theta}_{m_3} + D_{m_3} \dot{\theta}_{m_3} = \tau_{m_3}(t) + [J_{O_3} + J_{O_4} m r \omega^2] \ddot{\theta}_3 + [D_3 + D_4] \dot{\theta}_3$$

$$\begin{aligned} \tau_{m_3}(t) &= J_{m_3} \ddot{\theta}_{m_3} + D_{m_3} \dot{\theta}_{m_3} \\ & - [J_{O_3} + J_{O_4} m r \omega^2] \ddot{\theta}_3 \\ & - [D_3 + D_4] \dot{\theta}_3 \end{aligned}$$

$\therefore$ EOM₃ :-

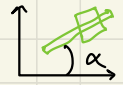
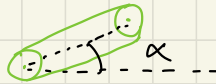
$$\frac{R_a}{K_T} \left[J_{m_3} \dot{\theta}_{m_3} + D_{m_3} \dot{\theta}_{m_3} - [J_{O_3} + J_{O_4} m r \omega^2] \dot{\theta}_3 - [D_3 + D_4] \dot{\theta}_3 \right] + K_b \dot{\theta}_{m_3} = v_3(t)$$

3. Power Screw :-

TEE

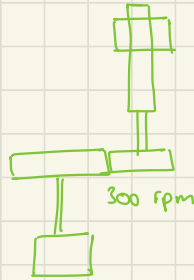
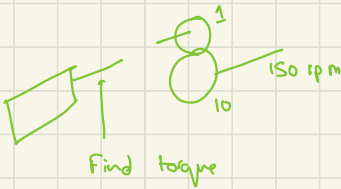
$$\text{Power} = \text{Torque} \times \Omega$$

1. Complete FBD / Component FBD
2. TB / PS DCM + GB



$$F = m \times \text{Acceleration} + C(\text{velocity})$$

↓
Damping



lead = 5 mm

5 mm → 1 rotation

